# Functional Near-Infrared Spectroscopy in Neonatal Research

Experimental design, data analysis and real-data preprocessing with Python

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# About us

## NeuroDevelopment and Comparative Cognition (NeuroDevCo)

*HORIZON-ERC-2023-StG: GALA (Gates to Language)*

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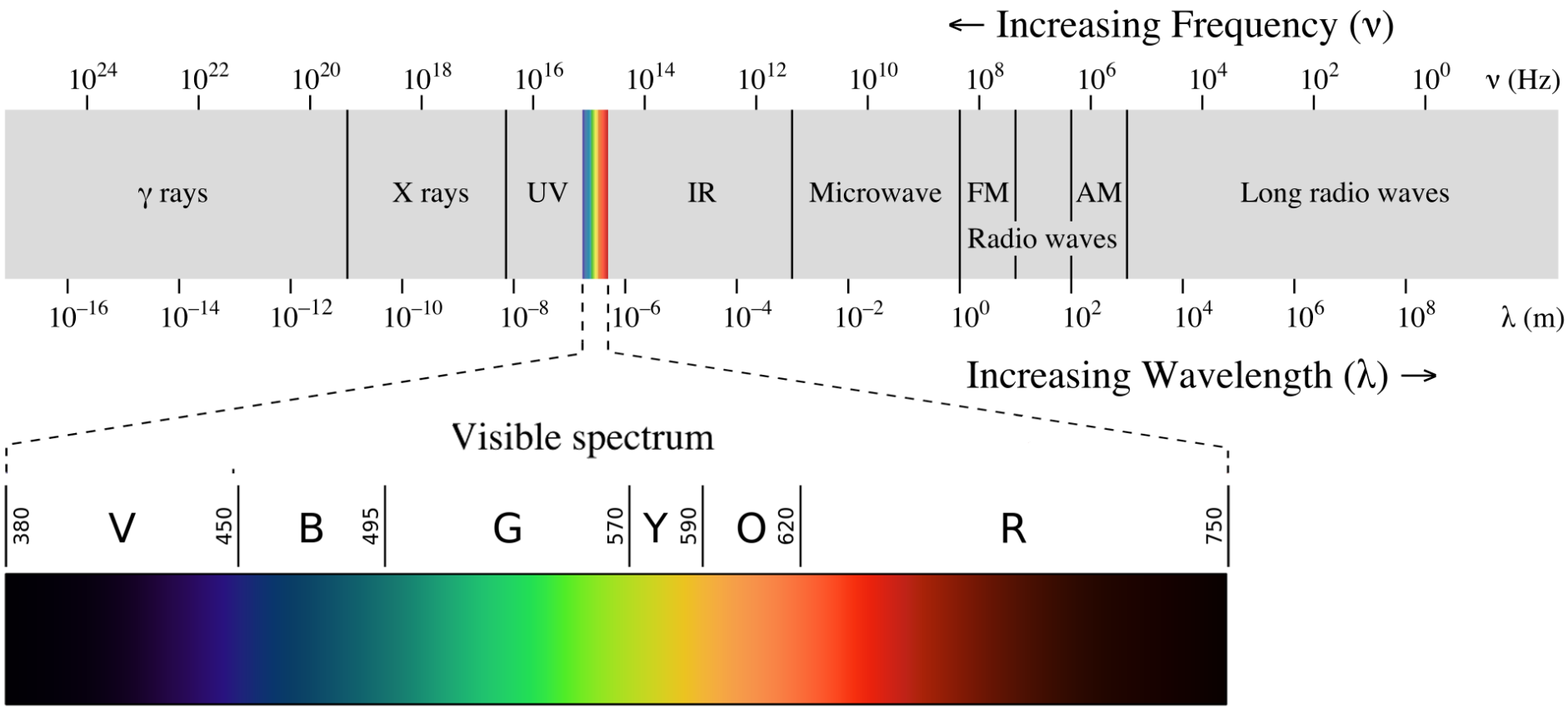
# Functional Near-Infrared Spectroscopy

# Functional Near-Infrared Spectroscopy

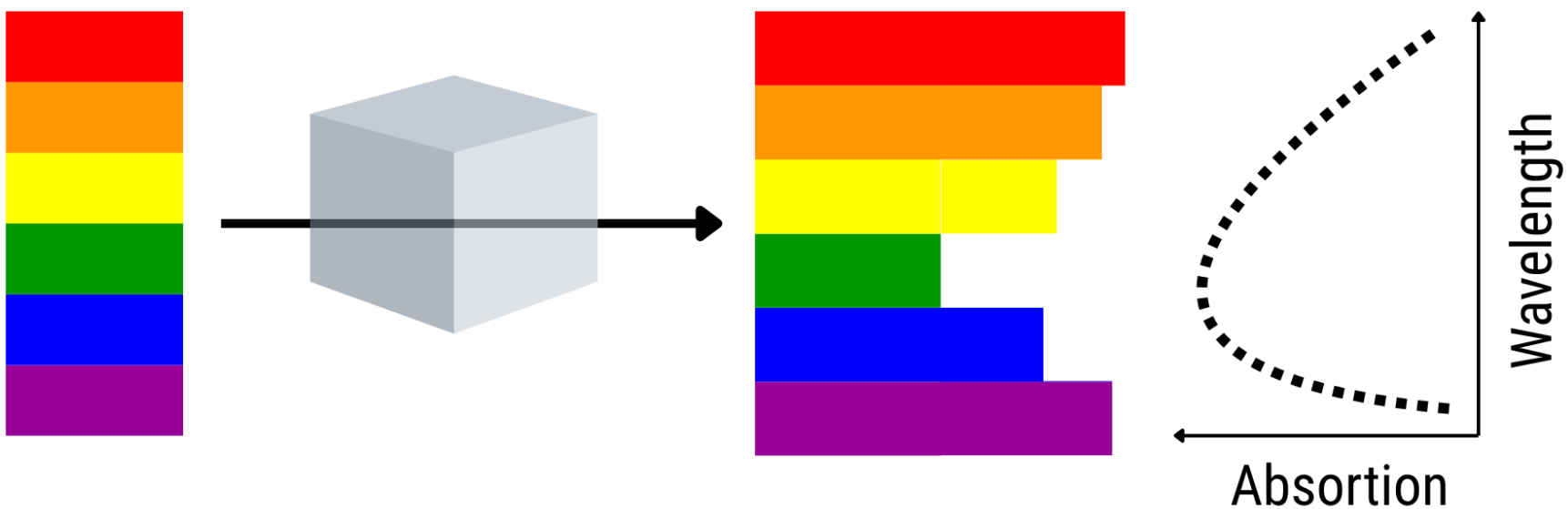
# Functional Near-Infrared Spectroscopy

Some materials have **different optical properties** than others (e.g., they absorbe more light of some colors).



# Functional Near-Infrared Spectroscopy

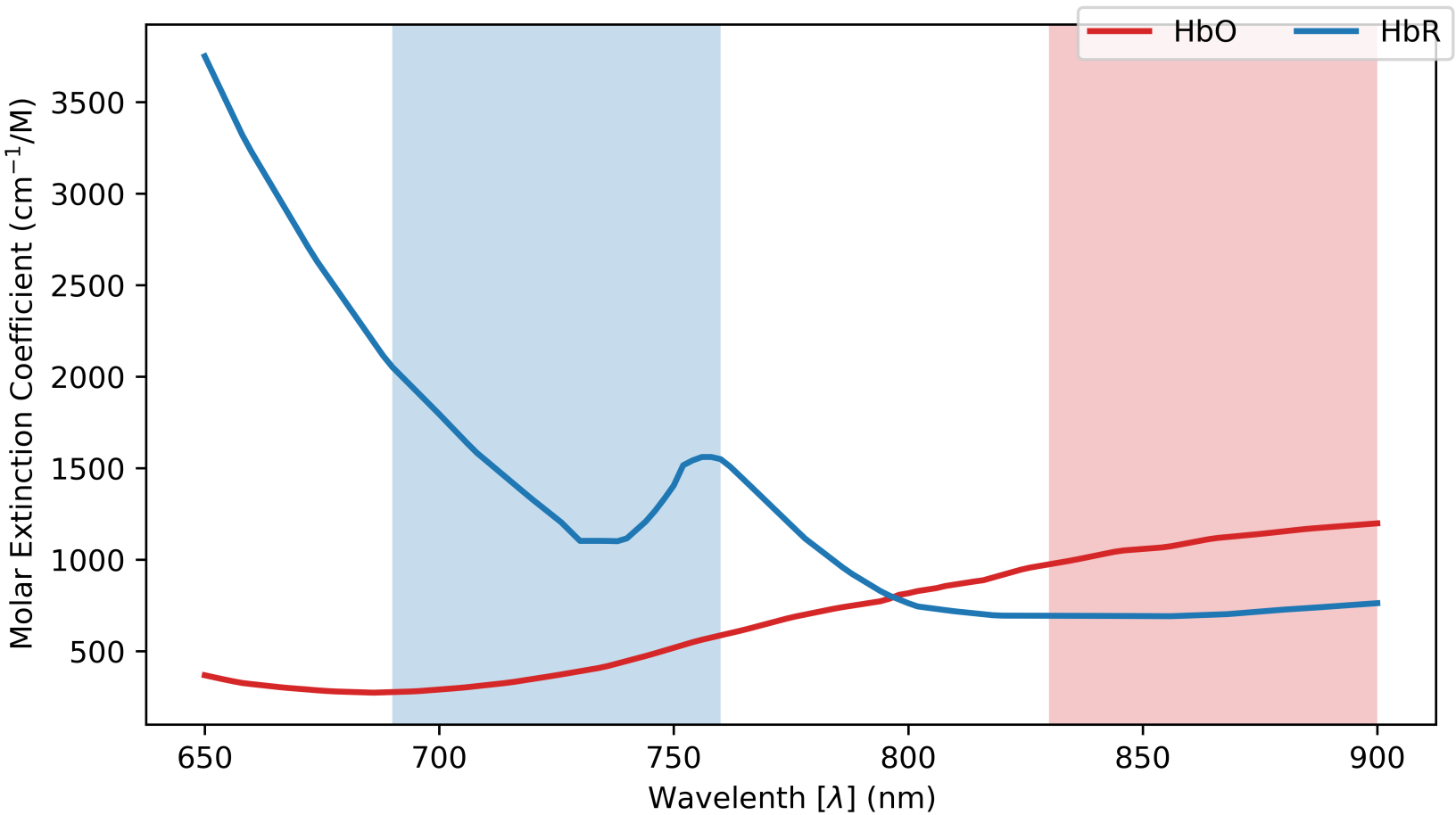
We can study the **composition of an object** by analysing how light behaves when passing through it (i.e., **spectroscopy**)



# Functional Near-Infrared Spectroscopy

# Functional Near-Infrared Spectroscopy

**Oxy-haemoglobin** (HbO) and **deoxy-haemoglobin** (HbR) absorb **near-infrared light** differently




NIR light absorption by Hb

# Functional Near-Infrared Spectroscopy

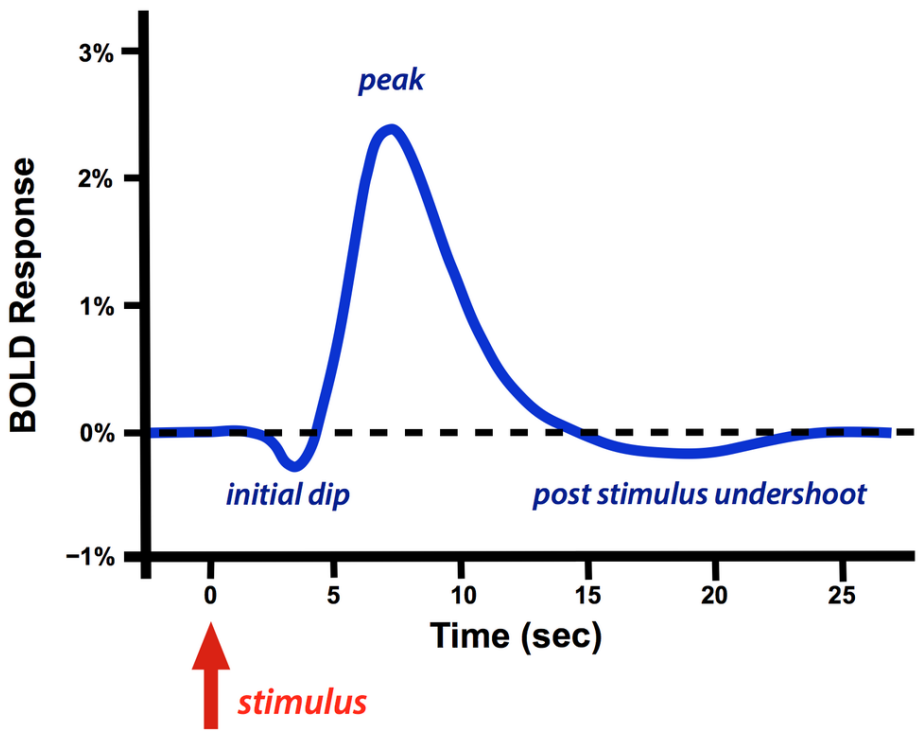
If we shine near-infrared light into biological tissue, we can estimate **changes in relative concentration** of HbO and HbR time

**modified Beer-Lambert Law (mBLL)**

- Describes **changes in light absorption** of underlying tissue using continuous waves methods
- Based on a generalization of **Beer-Lambert law**
- Assumes that the scattering properties of tissue do not vary with time

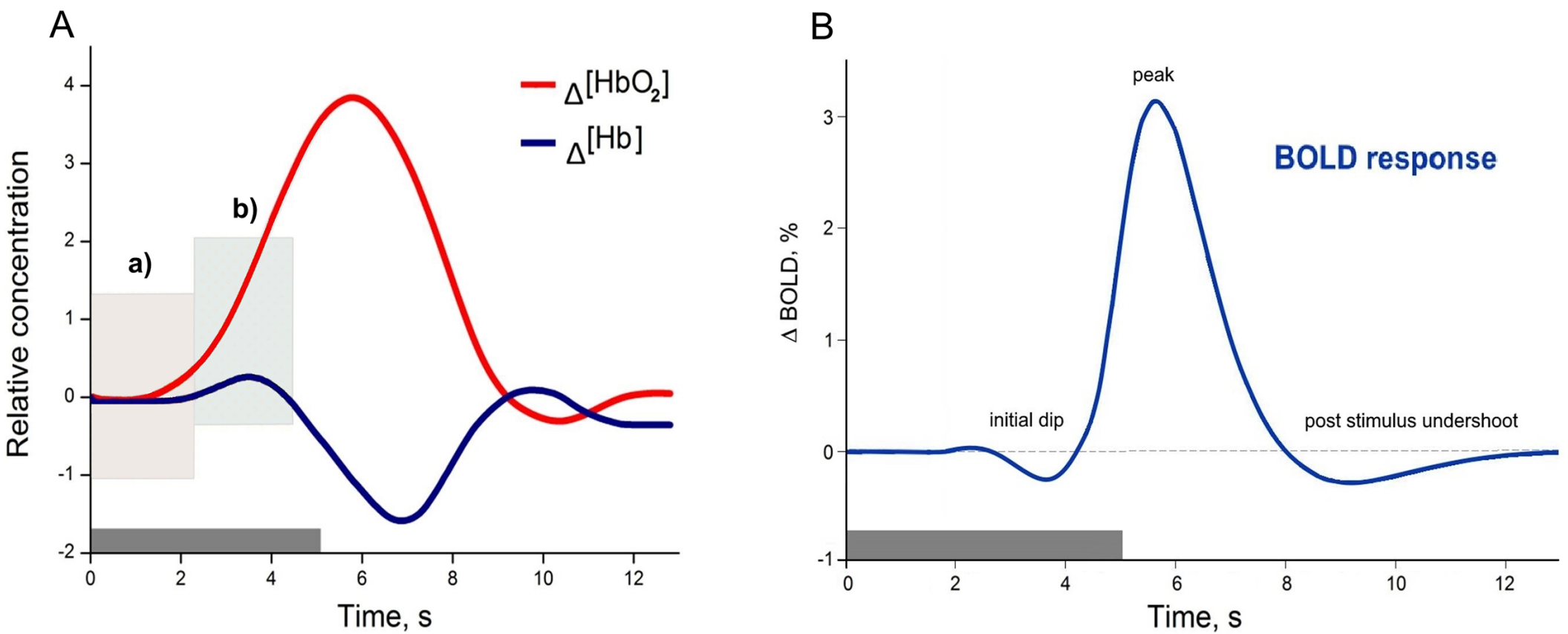
# Functional Near-Infrared Spectroscopy

- Blood-Oxygen-Level-Dependent (**BOLD**) response
- **Surplus of** HbO sent to recently active brain areas
- Takes ~15 seconds recover baseline



# Functional Near-Infrared Spectroscopy

## Canonical Haemodynamic Response Function (HRF)



# Resources

## The fNIRS Glossary project (Stute et al. 2025)

https://openfnirs.org/standards/fnirs-glossary-project/

Home Hardware Software Data Standards Trainings / Conferences Forum Login

About

FNIRS GLOSSARY PROJECT

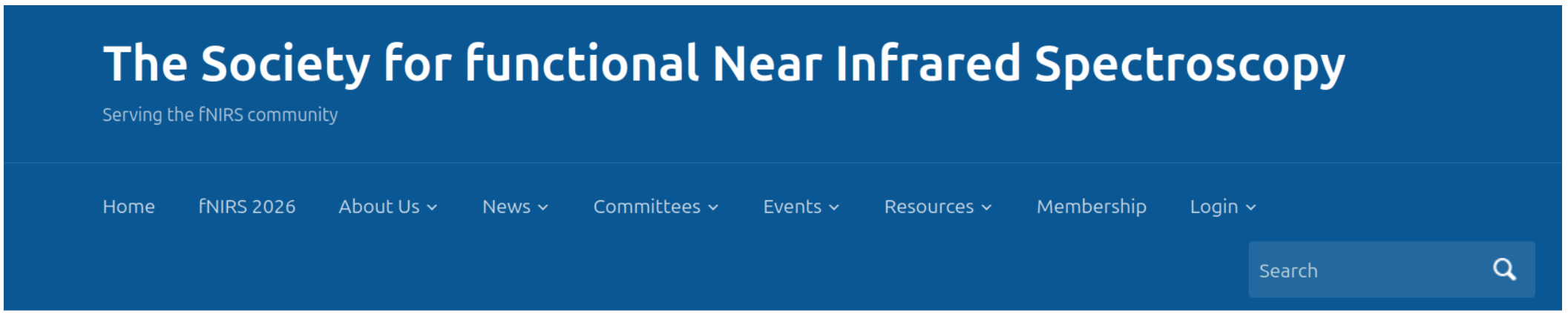
**The fNIRS Glossary Project** is a community-sourced glossary (i.e., useful nomenclature and related symbols) of terms related to fNIRS technology. The glossary includes terms related to continuous-wave (CW), frequency domain (FD), and time domain (TD) NIRS systems, covering the experimental design, pre-processing, data analysis, neuroscience, physics and mathematical areas that are crucial to understanding and using fNIRS successfully and reliably in research, industry and clinical settings.

You can either browse the terms or use the search bar below to get more details about each term.

We have a Forum for discussion or feedback on the existing terms. If you're missing a term, please use this Google Form to submit it. Before submitting a new term, please use the search function to ensure that the term is missing and also has not been added as a synonym to an existing term.

# Resources

## Society for Functional Near Infrared Spectroscopy (SFNIRS)




### What's New at fNIRS.org

SfNIRS Elections 2026 – Call for Nominations

Research Webinar – 29th May 2026

SfNIRS Newsletter Spring 2026

Research Webinar – 10th April 2026

## Home

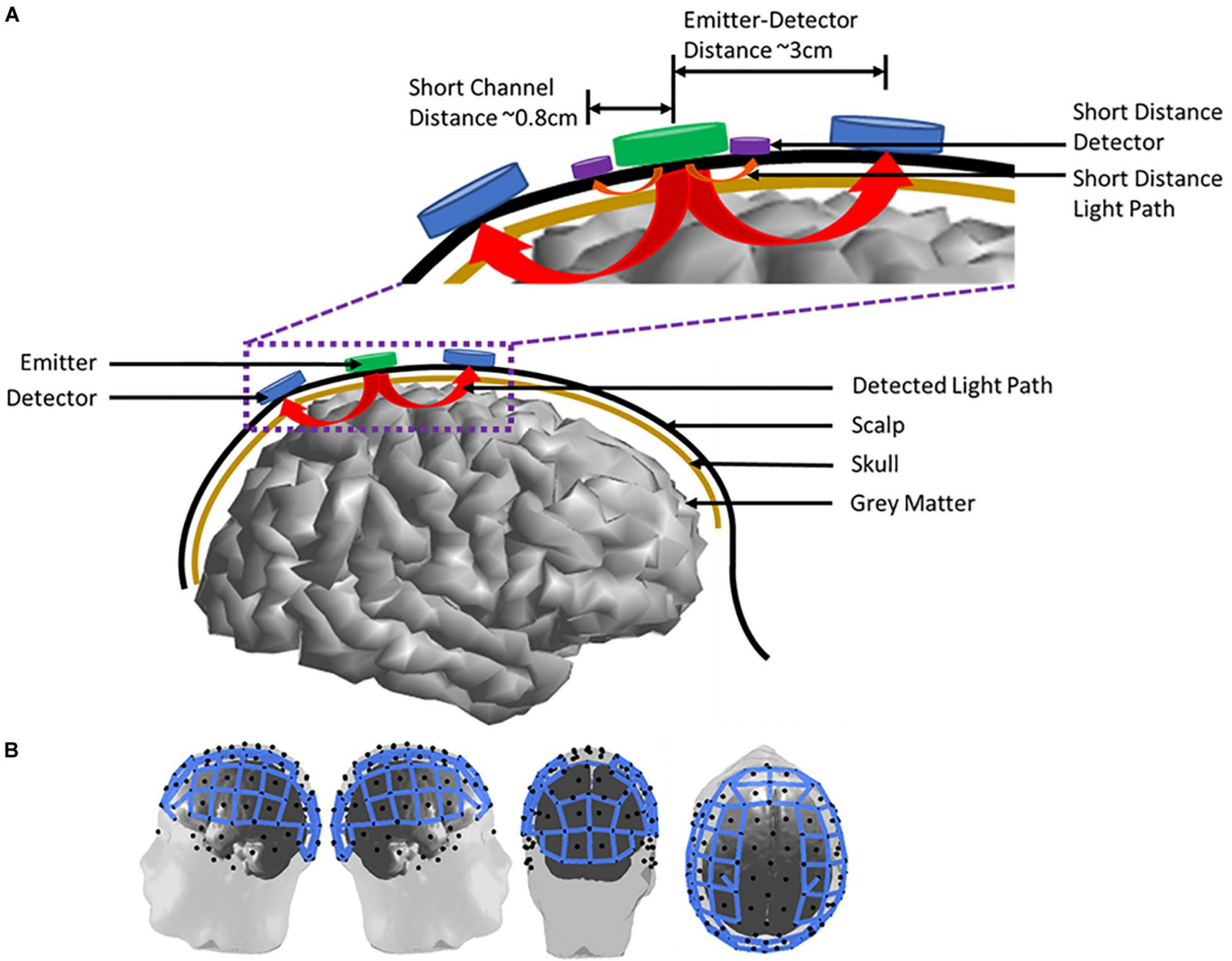
The Society for functional near-infrared spectroscopy (SfNIRS) is a professional organization of basic and clinical scientists who seek to understand the functional properties of biological tissues, especially the brain, using optical methods. The aim of the Society is to promote the exchange of ideas, interdisciplinary collaboration, and education.

Become a member

### Vision

A global home for functional near-infrared spectroscopy (fNIRS) serving both the scientific community and the public, fostering fNIRS as a high-performance, naturalistic, neuroimaging modality that drives transformative scientific research with high societal impact.

# fNIRS for experimental tasks



# fNIRS for experimental tasks

## Advantages

- Commercially available (still pricey)
- Relatively easy capping
- Flexible and portable, robust to movement*
- Hight-density diffuse optical tomography (HD-DOT) (Culver et al. 2001)

## Limitations

- Limited access to sub-cortical regions (~3 cm)*
- Temporal resolution constrained by haemodynamics
- Relatively pricey
- Limited spatial resolution

# Hardware

- NIRx (Germany): NIRSport2
- Artinis (The Netherlands): Brite, OxyMon
- Hitachi (Japan): ETG 4000
- GowerLabs (UK): LUMO (HD-DOT system)
- Cortivision (Poland): Spectrum C23 fNIRS
- PIONIRS (Italy): NIRSBOX

# fNIRS in neonates

# fNIRS at the Hospital Sant Joan de Déu

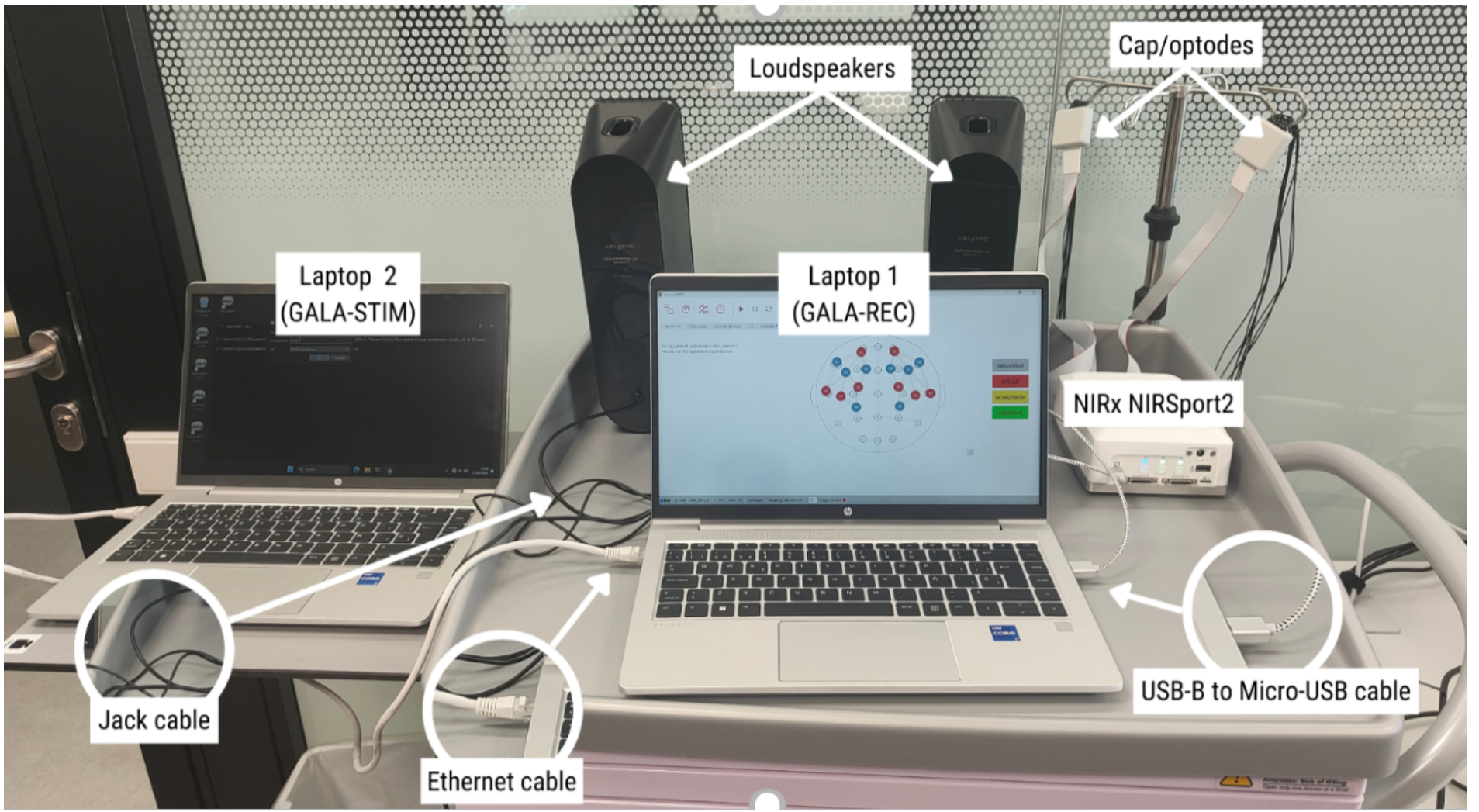
**Healthy** participants: ≥ 2,700 g & ≥ 8 Apgar-10 & ≥37 weeks post-mentrual age

Tested **6-72 hours after delivery**

**Cot-side testing** inside hospital room (Àrea de la Dona)

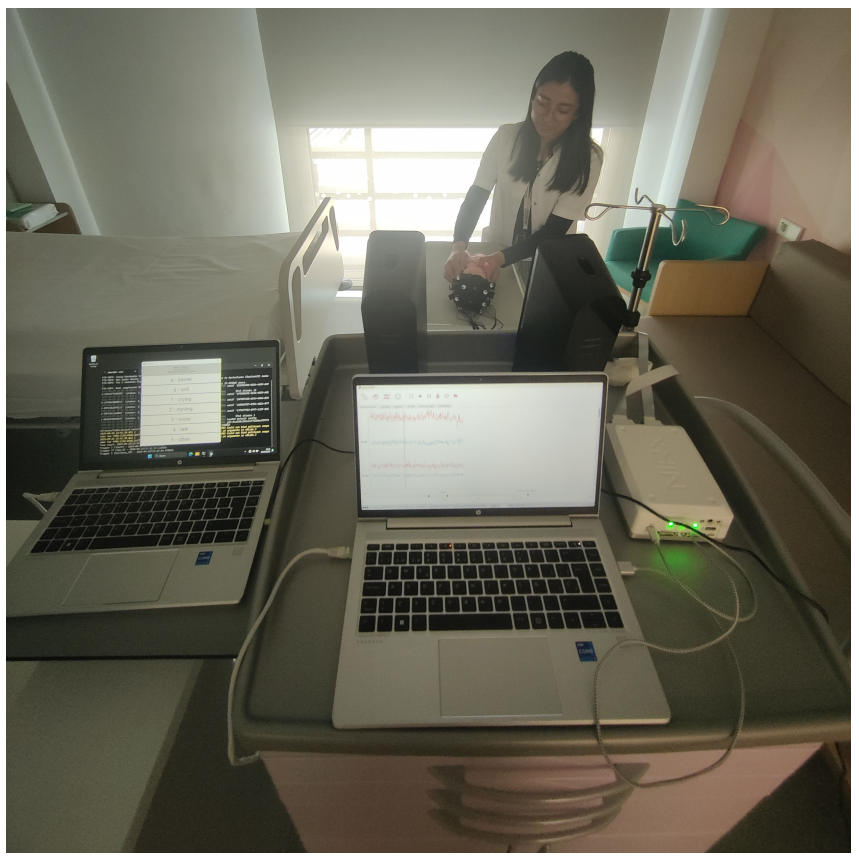
# fNIRS at the Hospital Sant Joan de Déu

## Setup



# fNIRS at the Hospital Sant Joan de Déu

## Setup



# fNIRS at the Hospital Sant Joan de Déu

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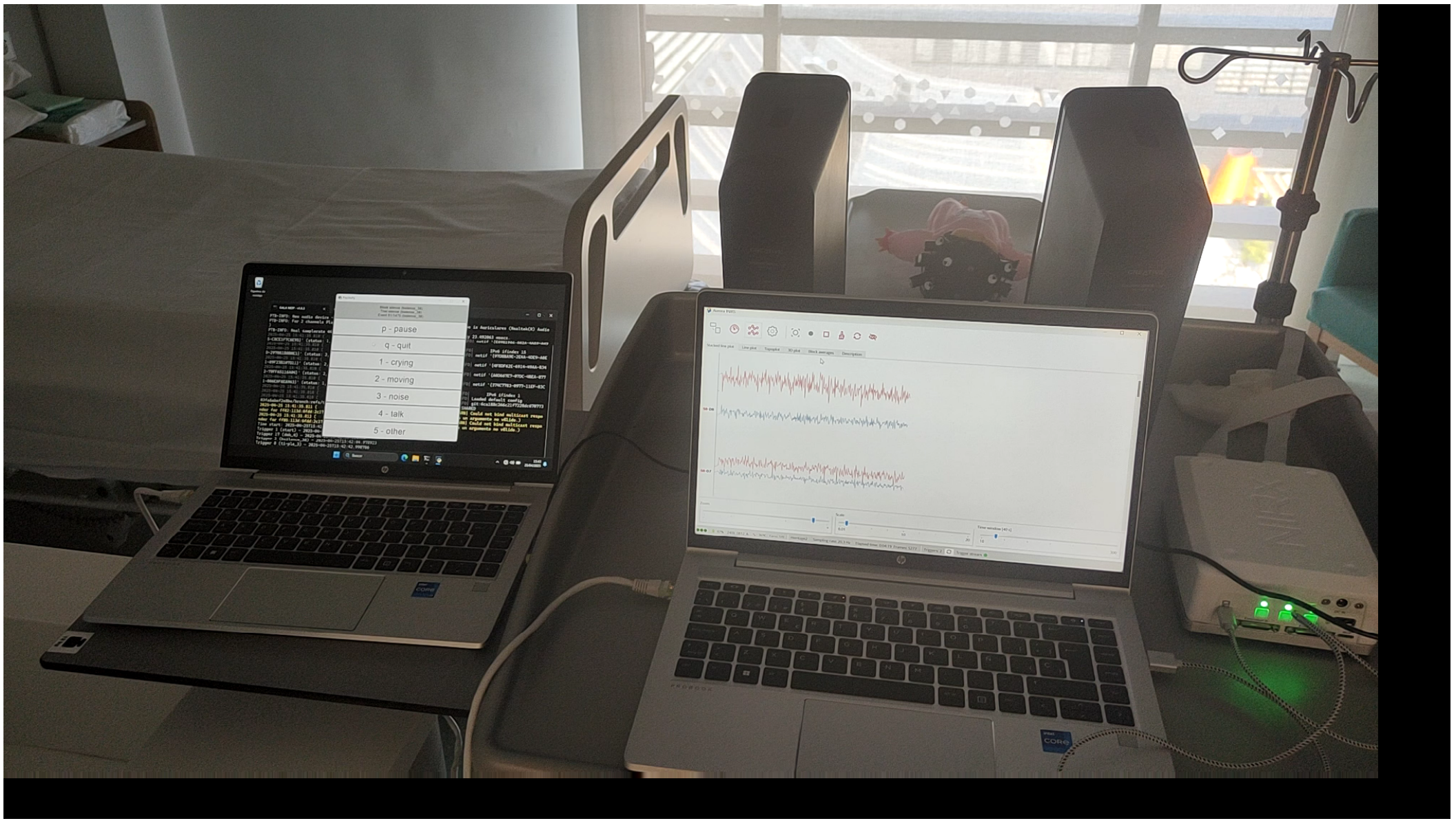
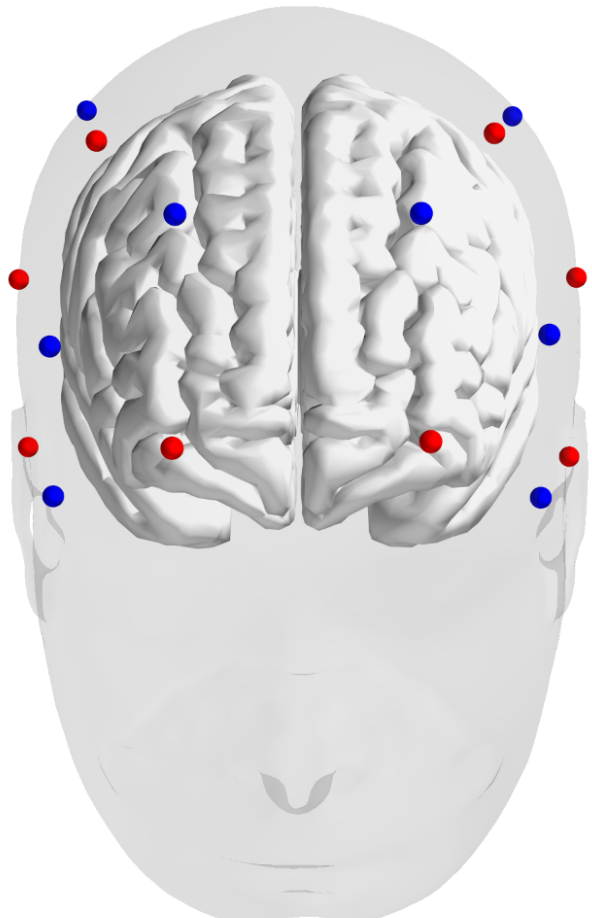
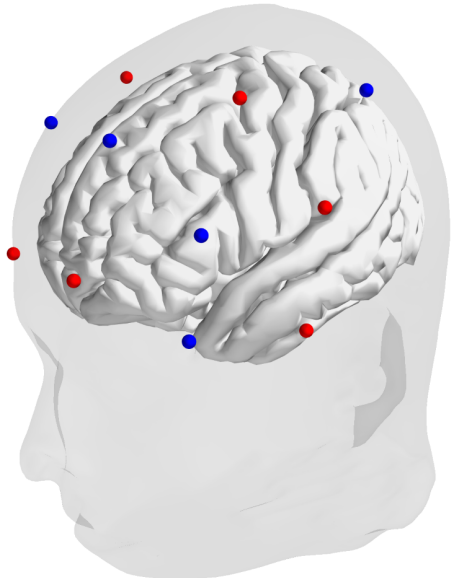
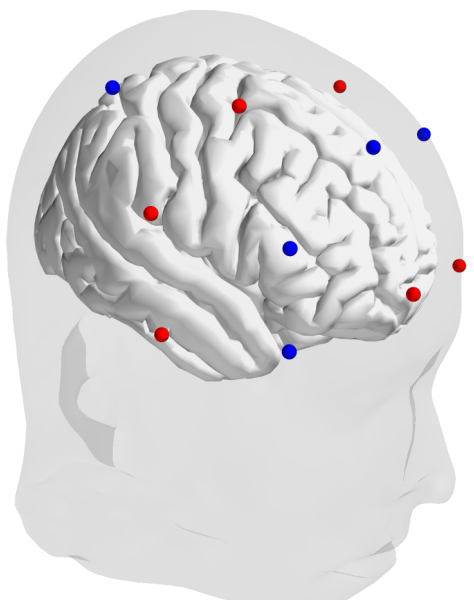


# fNIRS at the Hospital Sant Joan de Déu

## Montage/optode layout







# fNIRS at the Hospital Sant Joan de Déu

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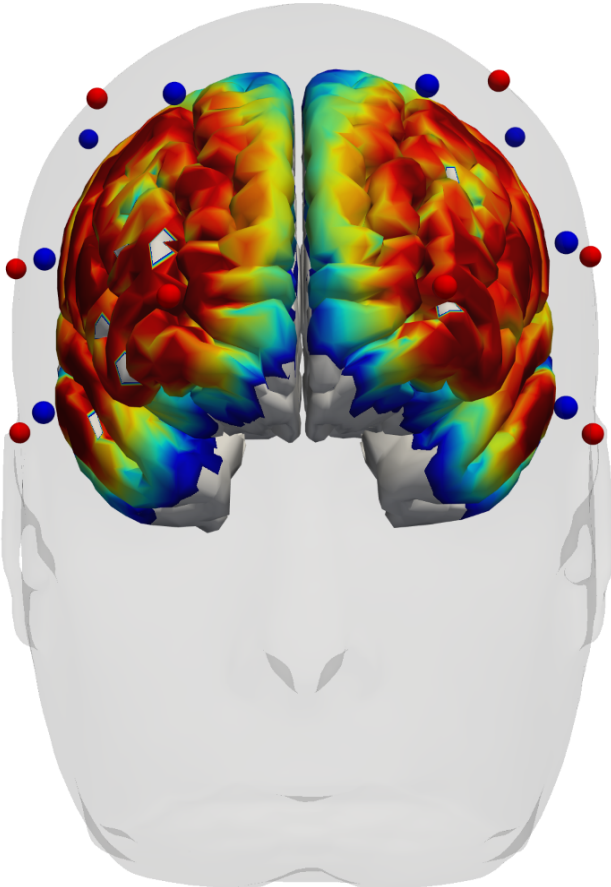
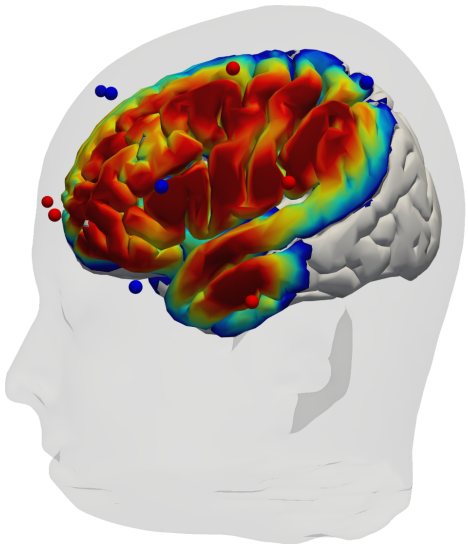
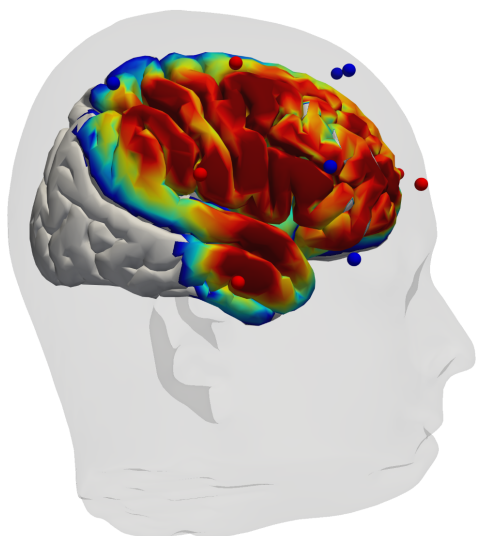




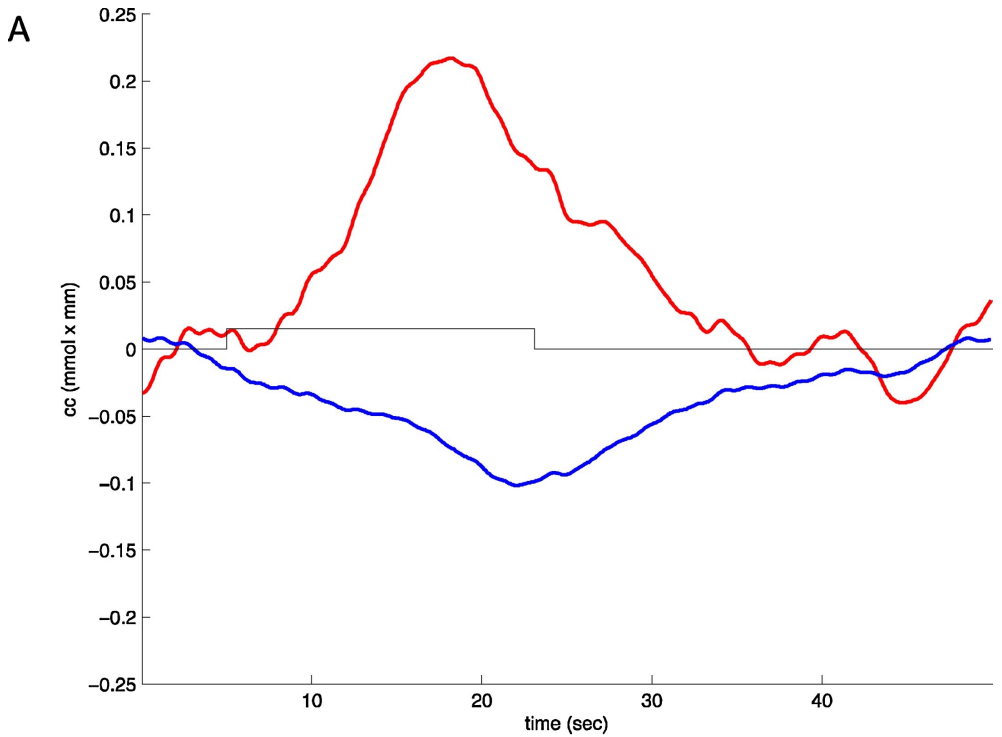


# Challenges in neonatal fNIRS

- True **shape of the HRF is relatively unknown** and variable within/between partiicpants and studies (Cristia et al. 2013; Lloyd-Fox, Blasi, and Elwell 2010)

# fNIRS in neonates

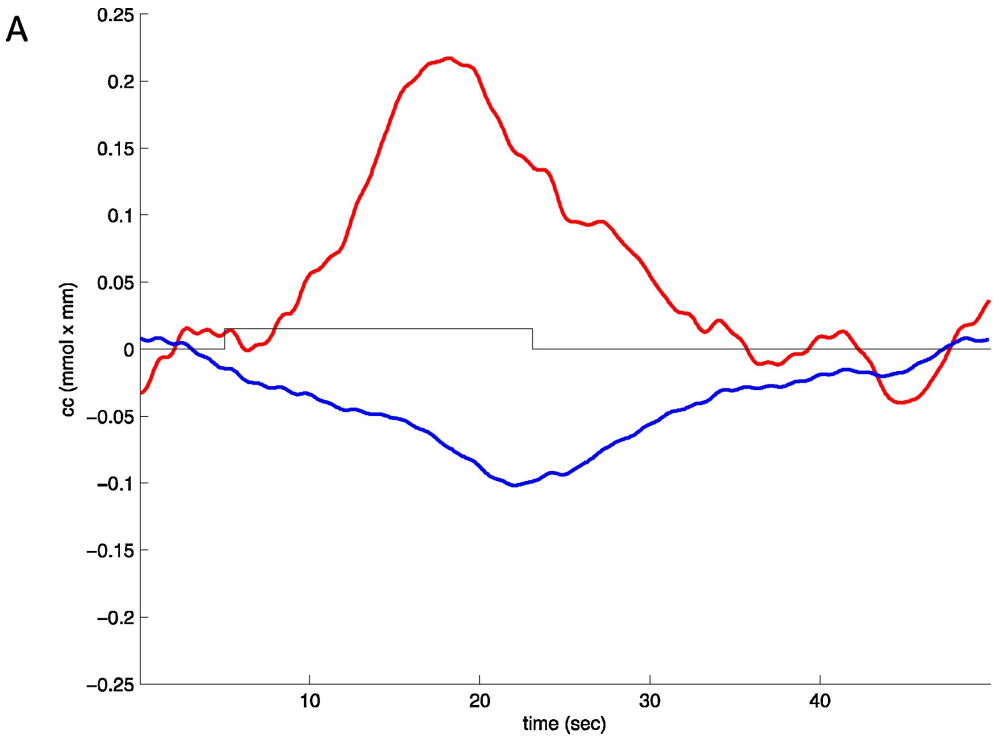
## Canonical response



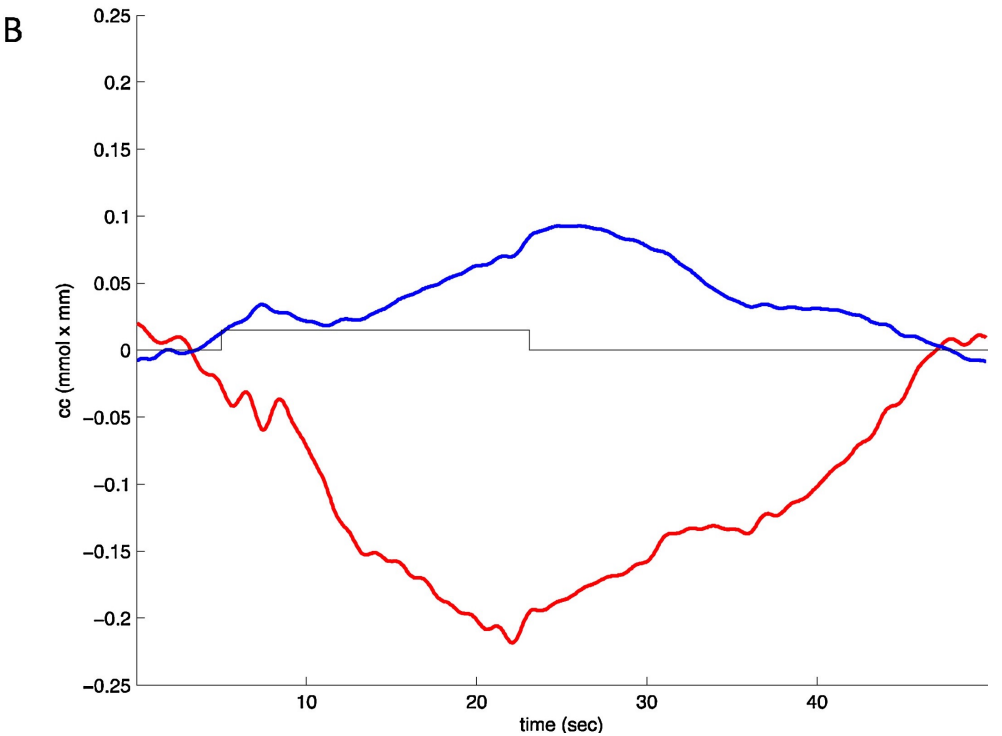

Issard and Gervain (2018)

# fNIRS in neonates

## Canonical response




## Inverted response




Issard and Gervain (2018)

# Challenges in neonatal fNIRS

- True **shape of the HRF is relatively unknown** and variable within/betweem partiicpants and studies (Cristia et al. 2013; Lloyd-Fox, Blasi, and Elwell 2010)
  - Ongoing maturation of the neurovascular coupling barrier? (Issard and Gervain 2018; Arichi et al. 2012)
- **Movement**, event during deep sleep
- **Wakefulness state**: ~40 min. sleep cycles
- **Head morphology**: spatial registration is challenging
- **Environment**: hospital ward

# Designing a preprocessing pipeline

## The FRESH initiative

Yücel, M. A., Luke, R., Mesquita, R. C., von Lühmann, A., Mehler, D. M., Lührs, M., … & Zemanek, V. (2025). **fNIRS reproducibility varies with data quality, analysis pipelines, and researcher experience**. *Communications biology*, 8(1), 1149.

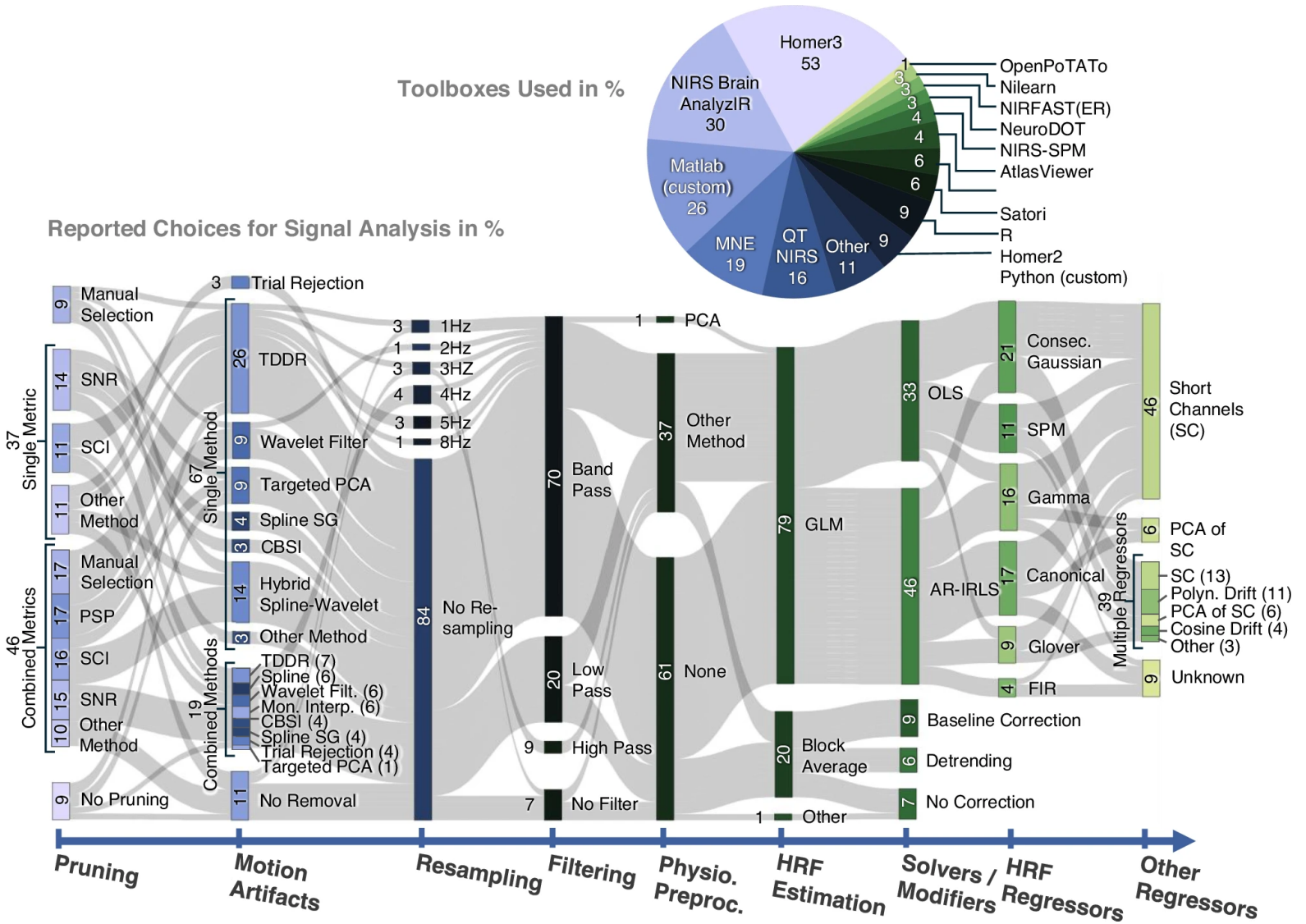
# Designing a preprocessing pipeline

## The FRESH initiative

- Asked **38 research teams** worldwide to independently analyze the same two fNIRS datasets.
- Using different pipelines, **nearly 80% of teams agreed on group-level results**
- Teams with higher self-reported analysis confidence, which correlated with years of fNIRS experience, showed greater agreement
- Main sources of variability were related to how poor-quality data were handled

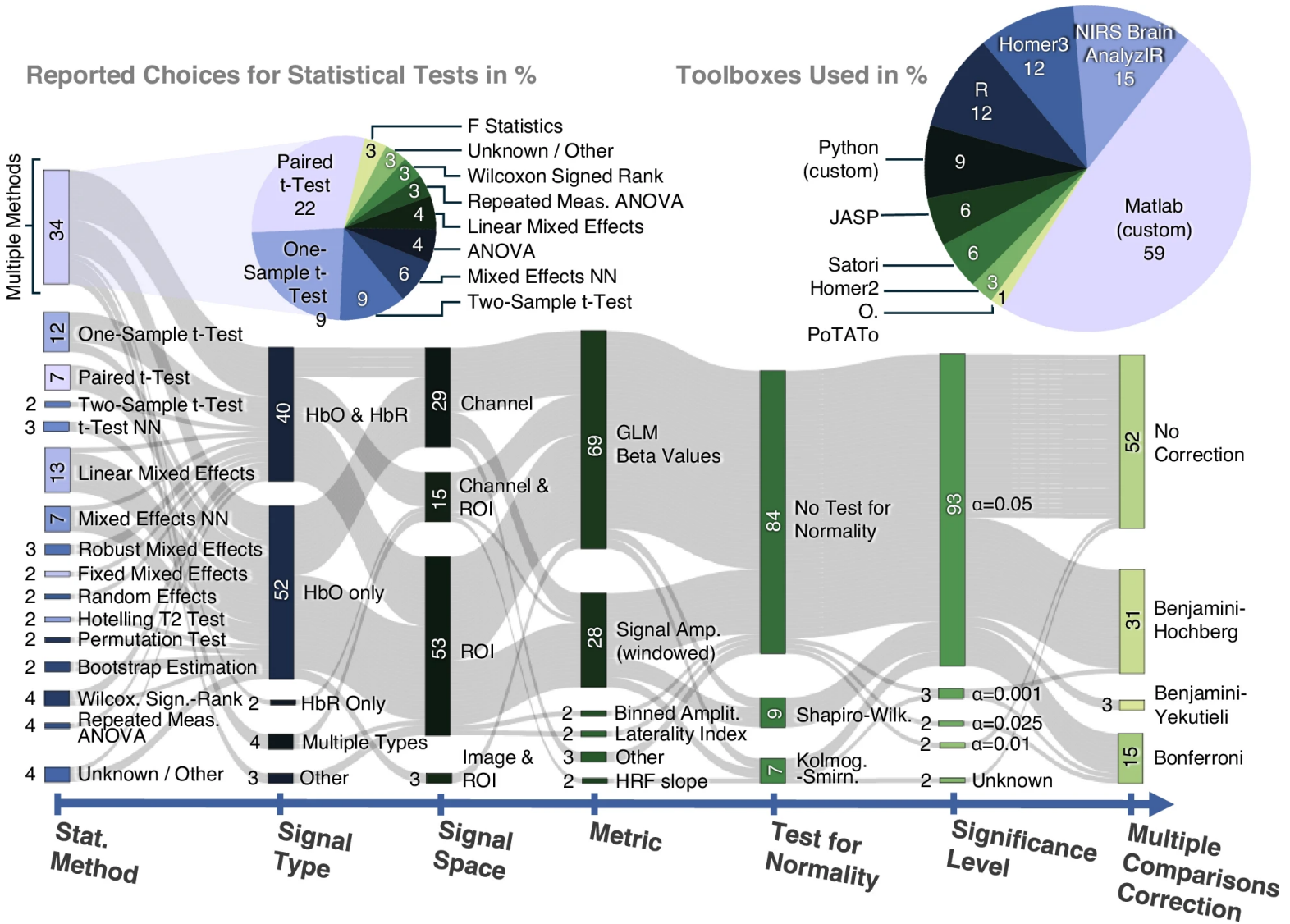
# Designing a preprocessing pipeline

## The FRESH initiative



# Designing a preprocessing pipeline

## The FRESH initiative



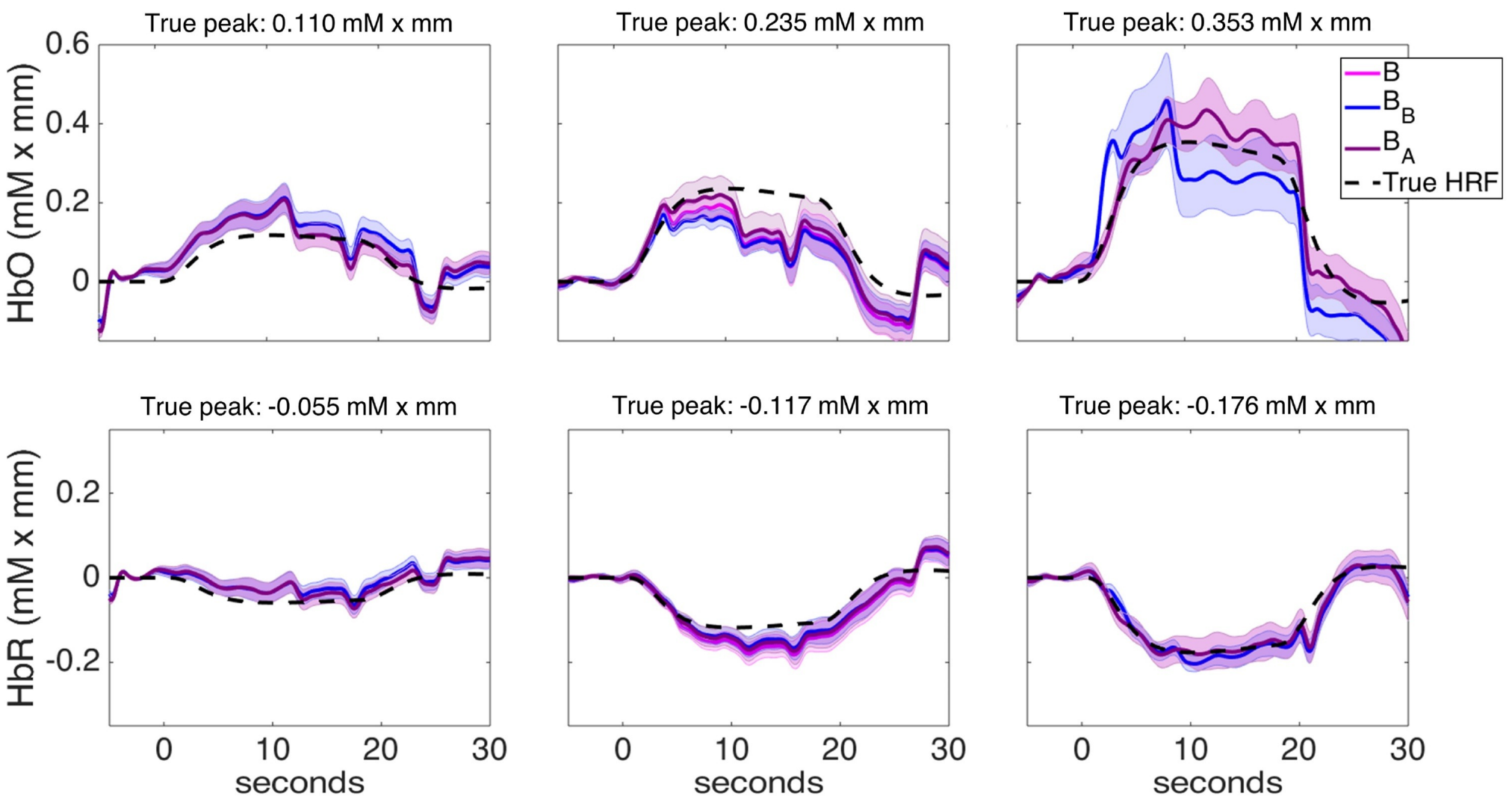
# Designing a preprocessing pipeline

Gemignani, J., & Gervain, J. (2021). **Comparing different pre-processing routines for infant fNIRS data**. *Developmental Cognitive Neuroscience*, 48, 100943.

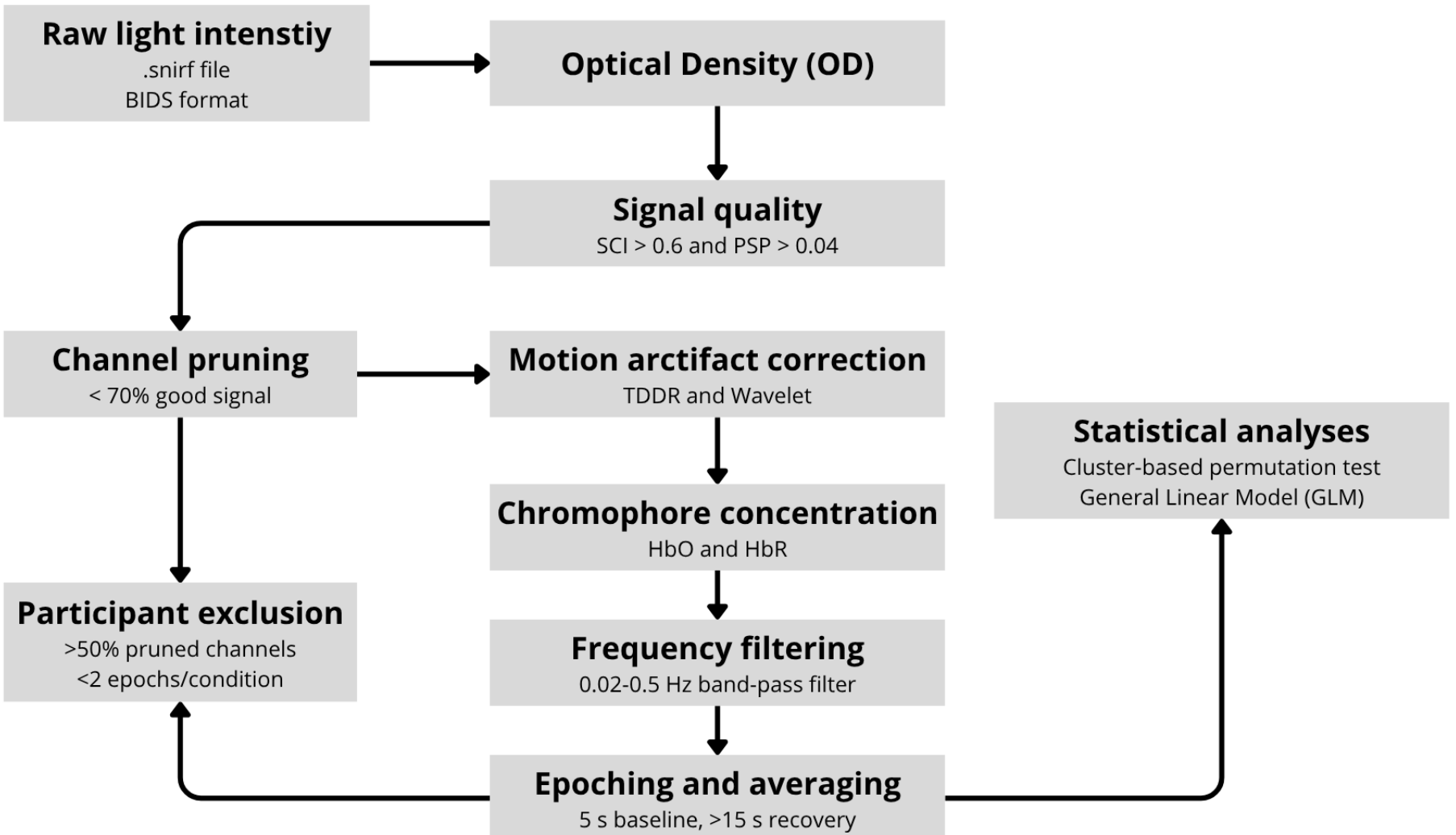
# Designing a preprocessing pipeline

- Data **quality/inclusion trade-off**
- Performance of all pipelines deteriorated with increasing levels of **noise**
- Use automatic motion artifact correction with **caution**: false negatives, reduced effect sizes, underestimation of the HRF
- The **ordering** of some preprocessing steps does not impact conclusions overall

# Designing a preprocessing pipeline



# Pipeline suggestion

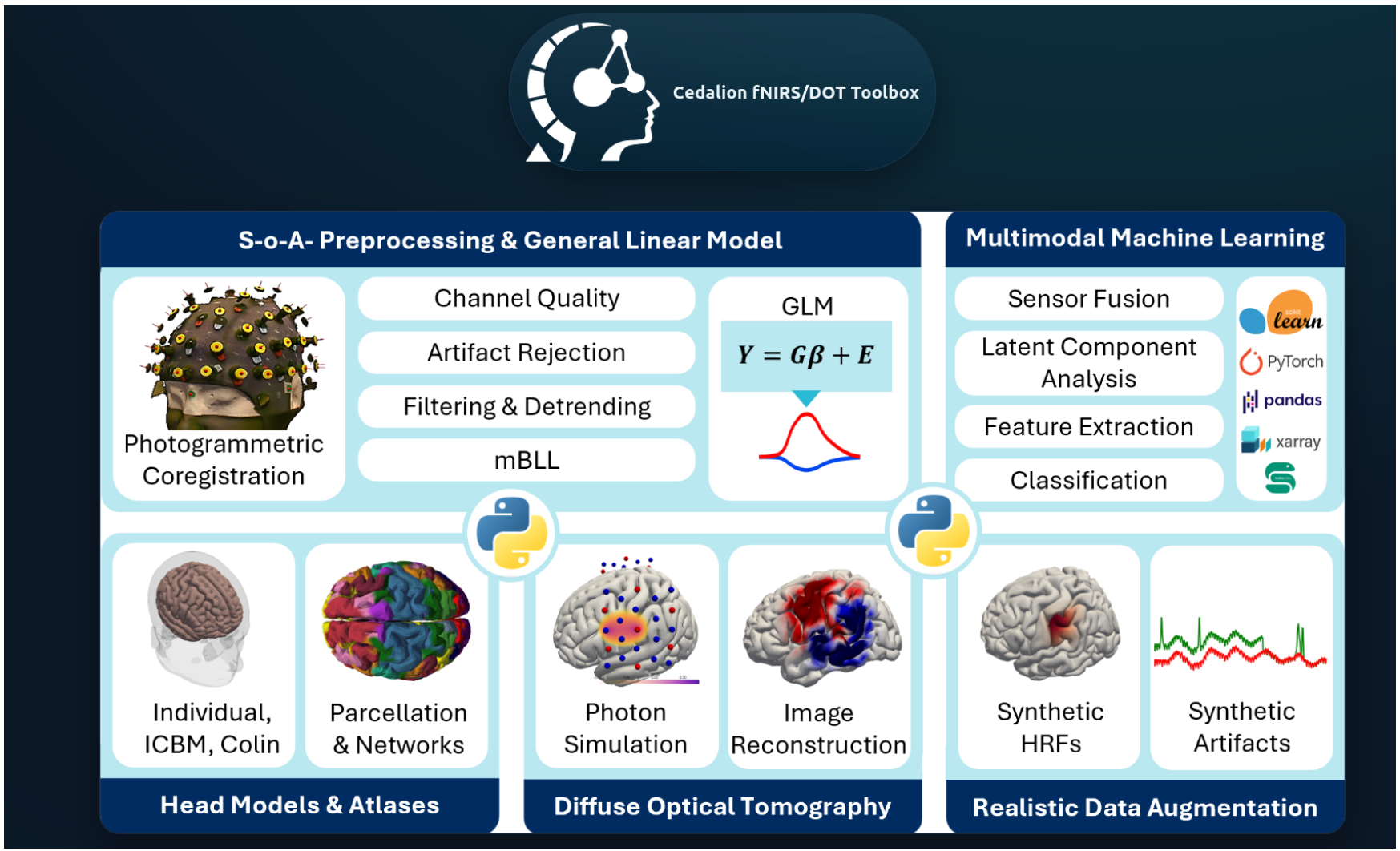


# Software

- Homer2/Homer3 (Huppert et al. 2009)
- MNE-Python (Gramfort et al. 2013), MNE-NIRS (Luke et al. 2021)
- AnalyzIR (Santosa et al. 2018)
- Brainstorm (Tadel et al. 2011), NIRStorm (Delaire et al. 2025)
- OxySoft (Artinis)
- NIRSTORM
- Fieldtrip
- Cedalion (Middell et al. 2026)

# Software

## Cedalion (Middell et al. 2026)



# Participant case study

# Datasets

## Neonates

García-Castro G., Sofocleous C., Gervain, J., Alarcón Allen A., and Santolin C. (in prep)

## Adults

Hernández-Sauret, A., García-Castro, G., & Redolar-Ripoll, E. (2026). *Brain Topography*, 39(1), 2. https://link.springer.com/article/10.1007/s10548-025-01157-4

# Datasets

50th ANNIVERSARY EDITION
CLINT EASTWOOD
THE GOOD
THE BAD
and THE UGLY
co-starring
LEE VAN CLEEF
also starring
ELI WALLACH
in the role of TUCO
directed by
SERGIO LEONE

# The good: adult data

Importing raw light intensity from `.snirf` file:

```
from pathlib import Path
from cedalion import io

rec = io.read_snirf(Path("data/adult.snirf"))[0]

print(rec)
```

- `rec["amp"]`: Raw light intensity dataset
- `rec["masks"]`: Signal quality masks
- `rec["stim"]`: Dataset of events (e.g., stimulus triggers, annotations)
- `rec["aux_ts"]`: Additional timeseries (e.g., accelerometer, heatbeat rate)

# The good: adult data

Plotting raw light intensity data with matplotlib:

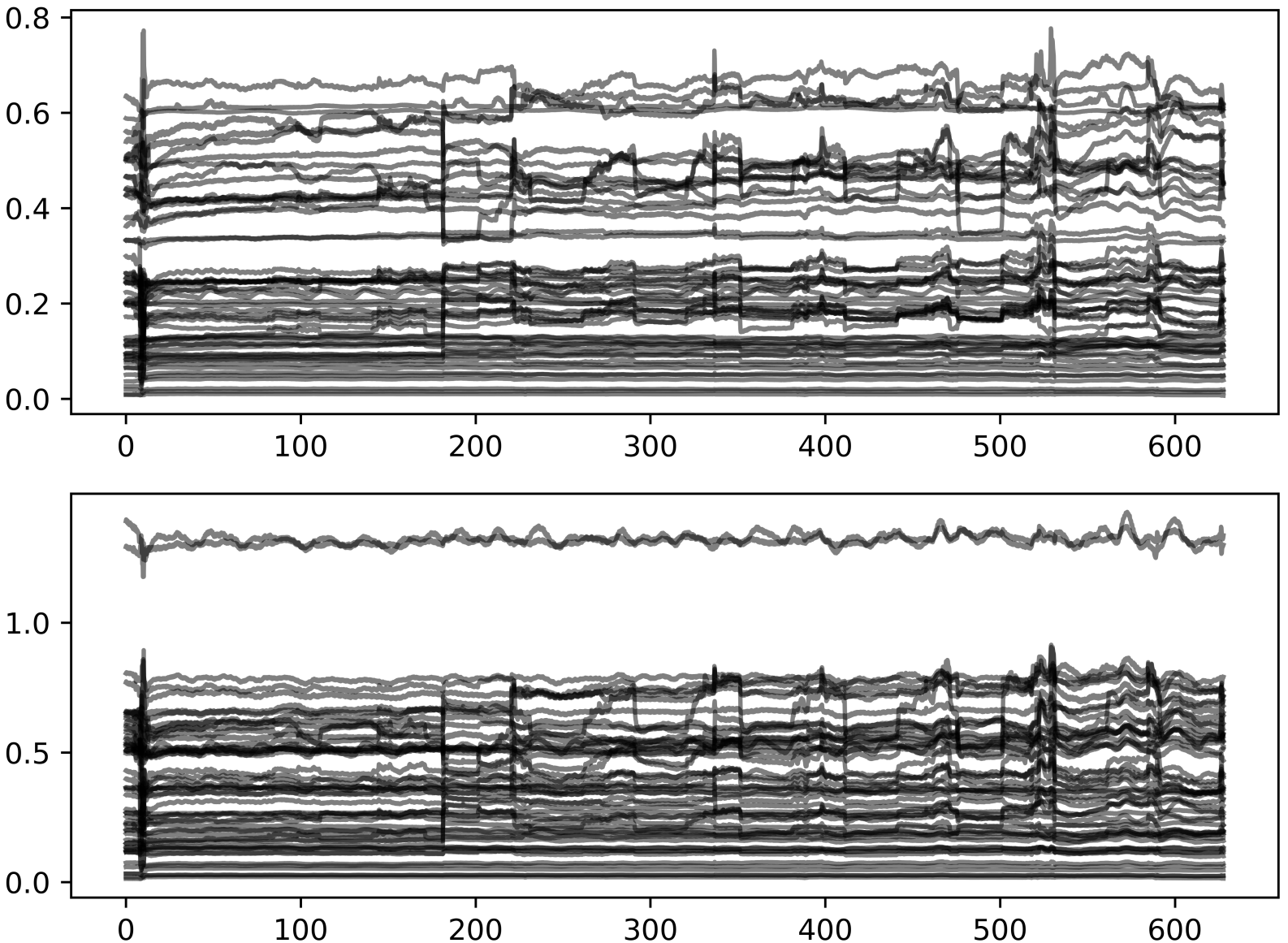
```
from matplotlib import pyplot as plt

amp = rec["amp"]

fig, axes = plt.subplots(2, 1)

for wl_i, wl in enumerate(amp.wavelength.values):
    for ch in amp.channel.values:
        d = amp.sel(wavelength=wl, channel=ch)
        axes[wl_i].plot(d.time, d, c="k", alpha=0.5)
```

# The good: adult data



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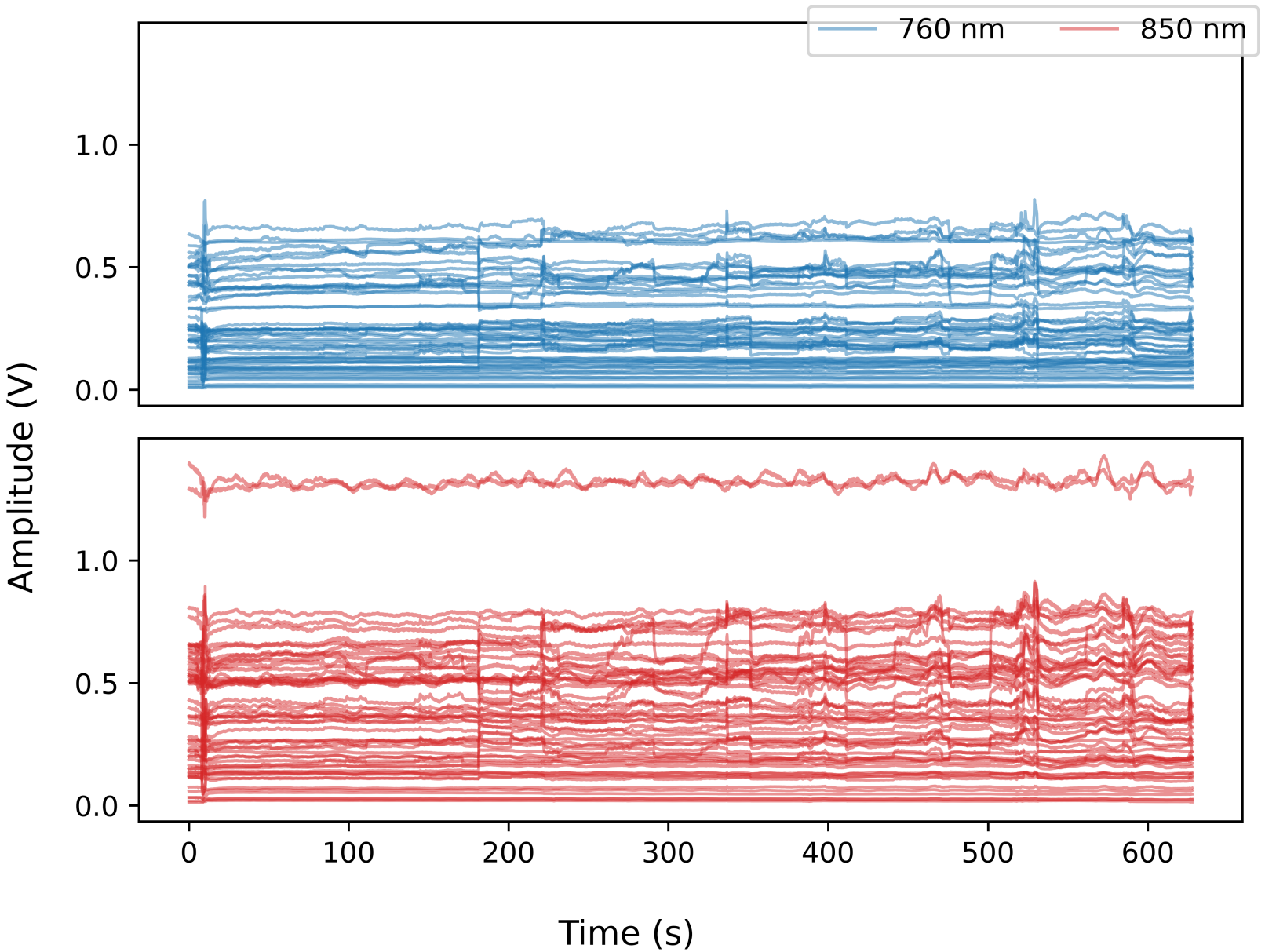

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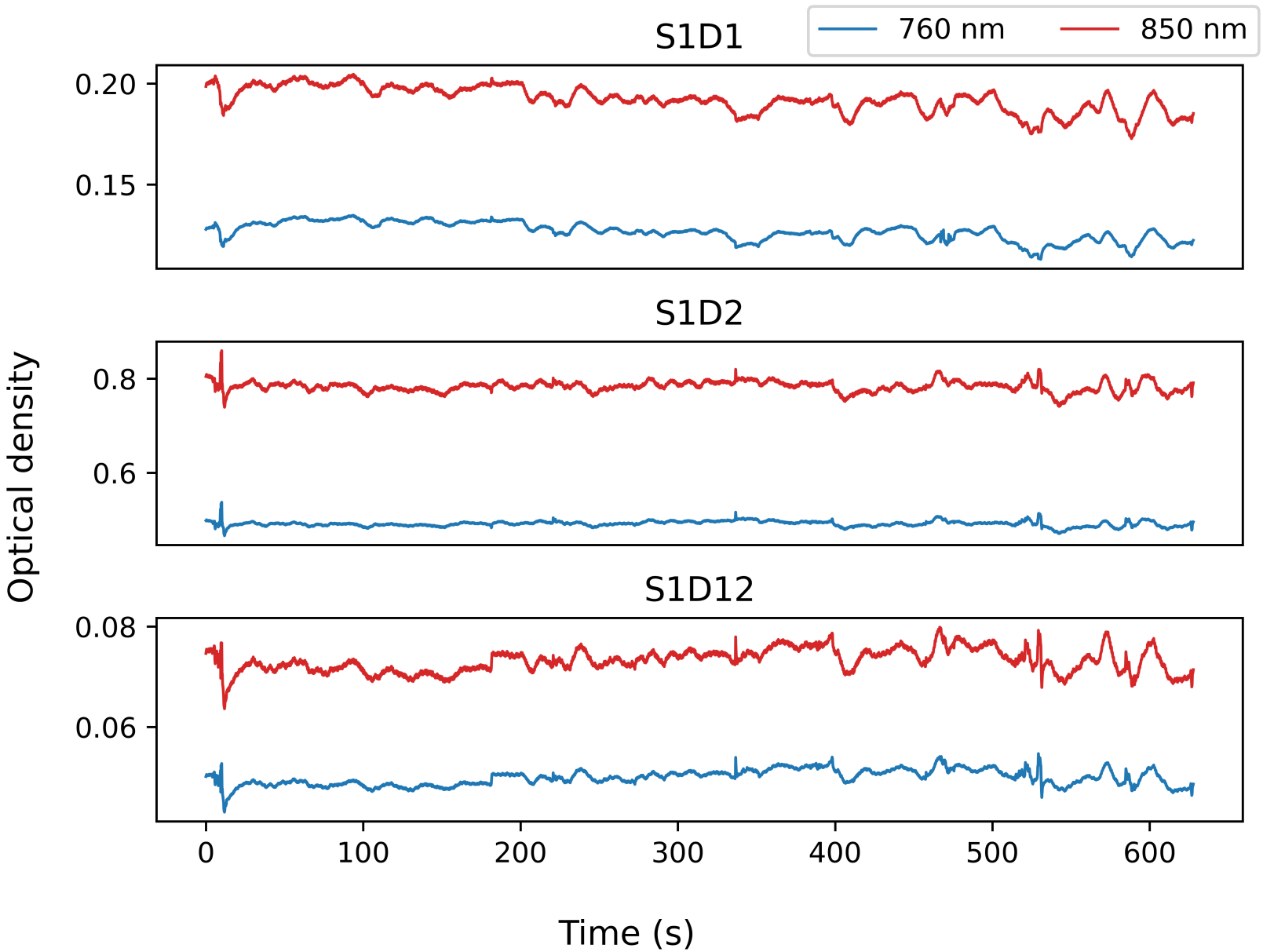
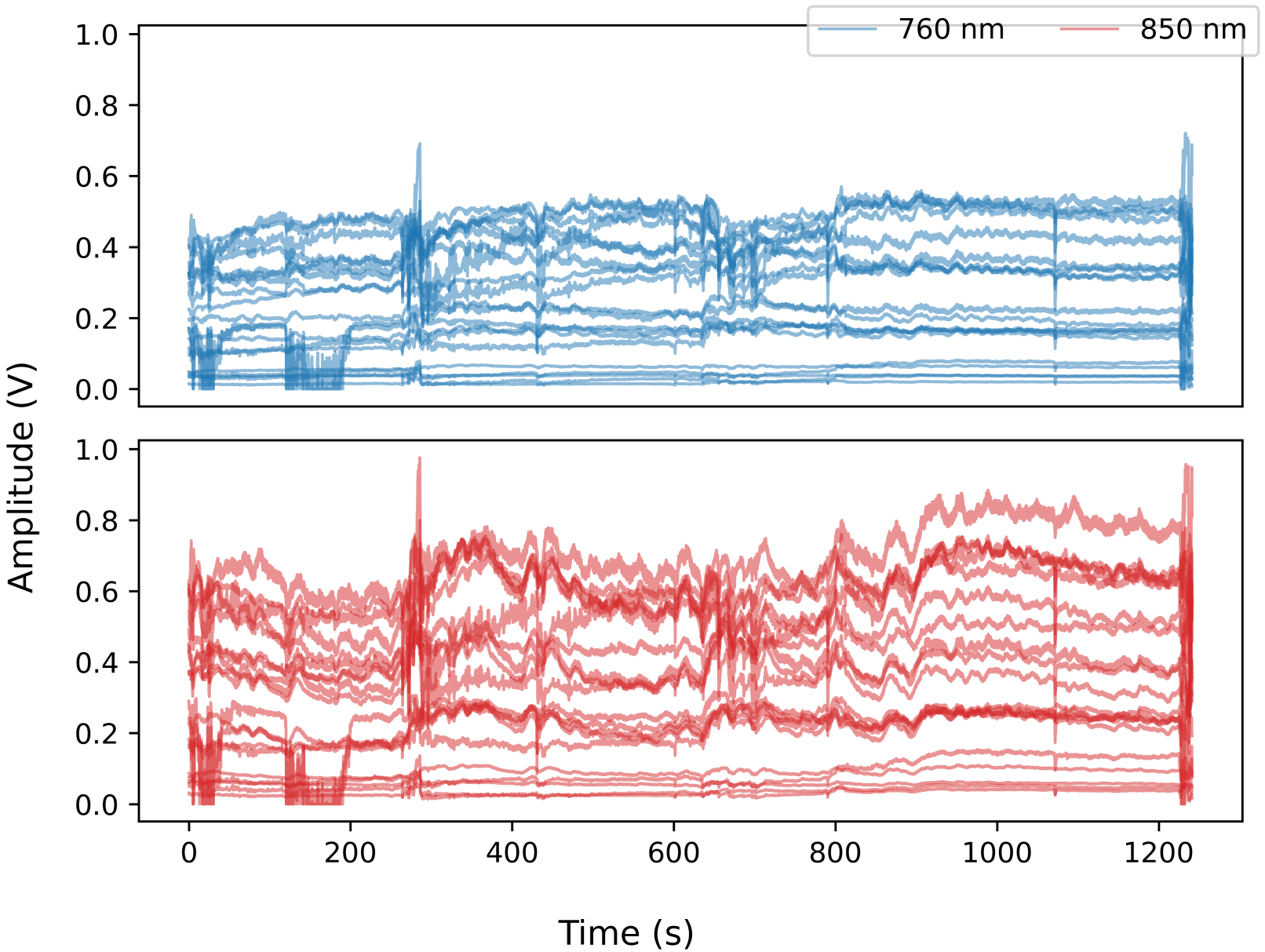


# The bad: breastfeeding neonate



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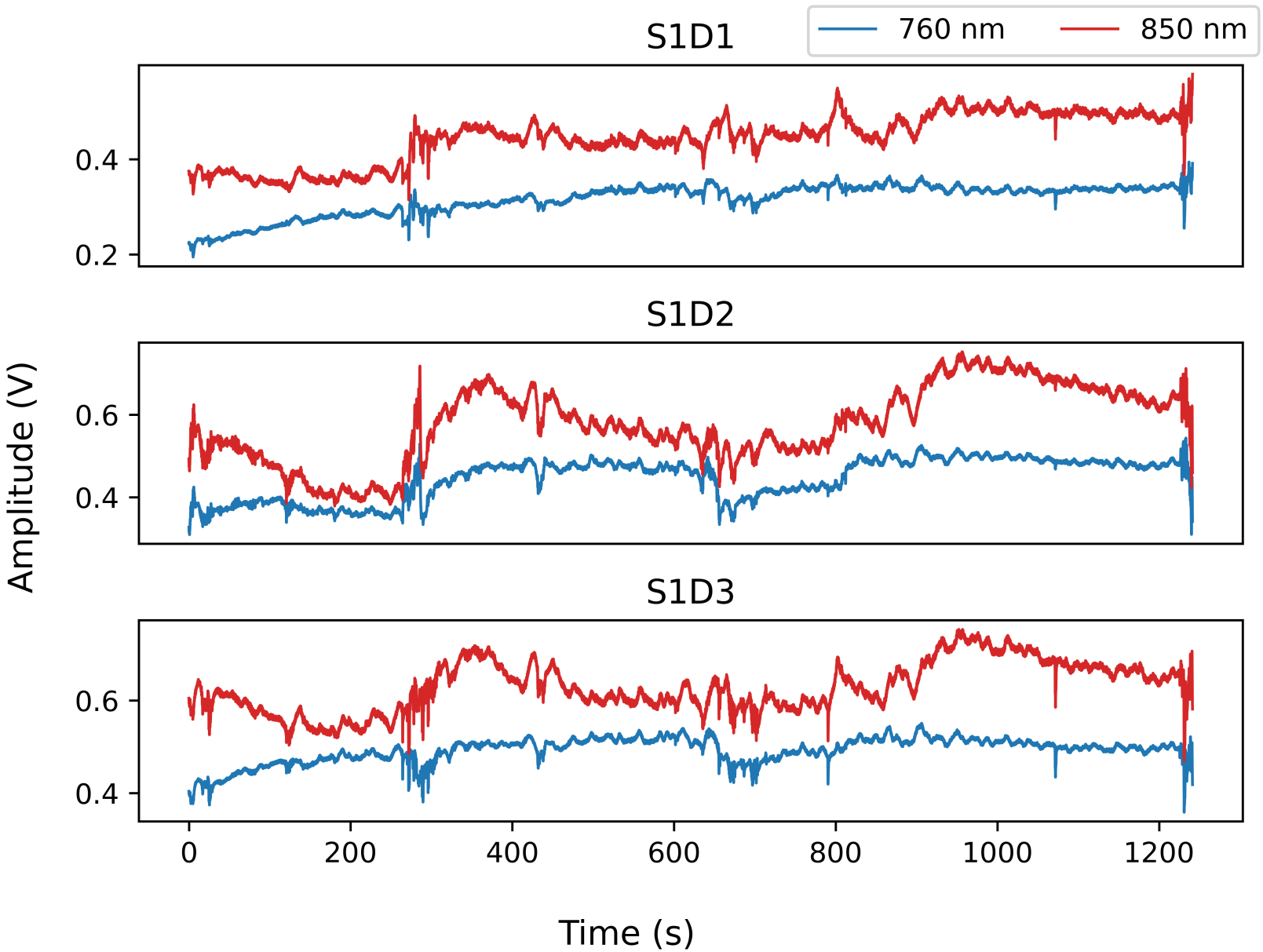

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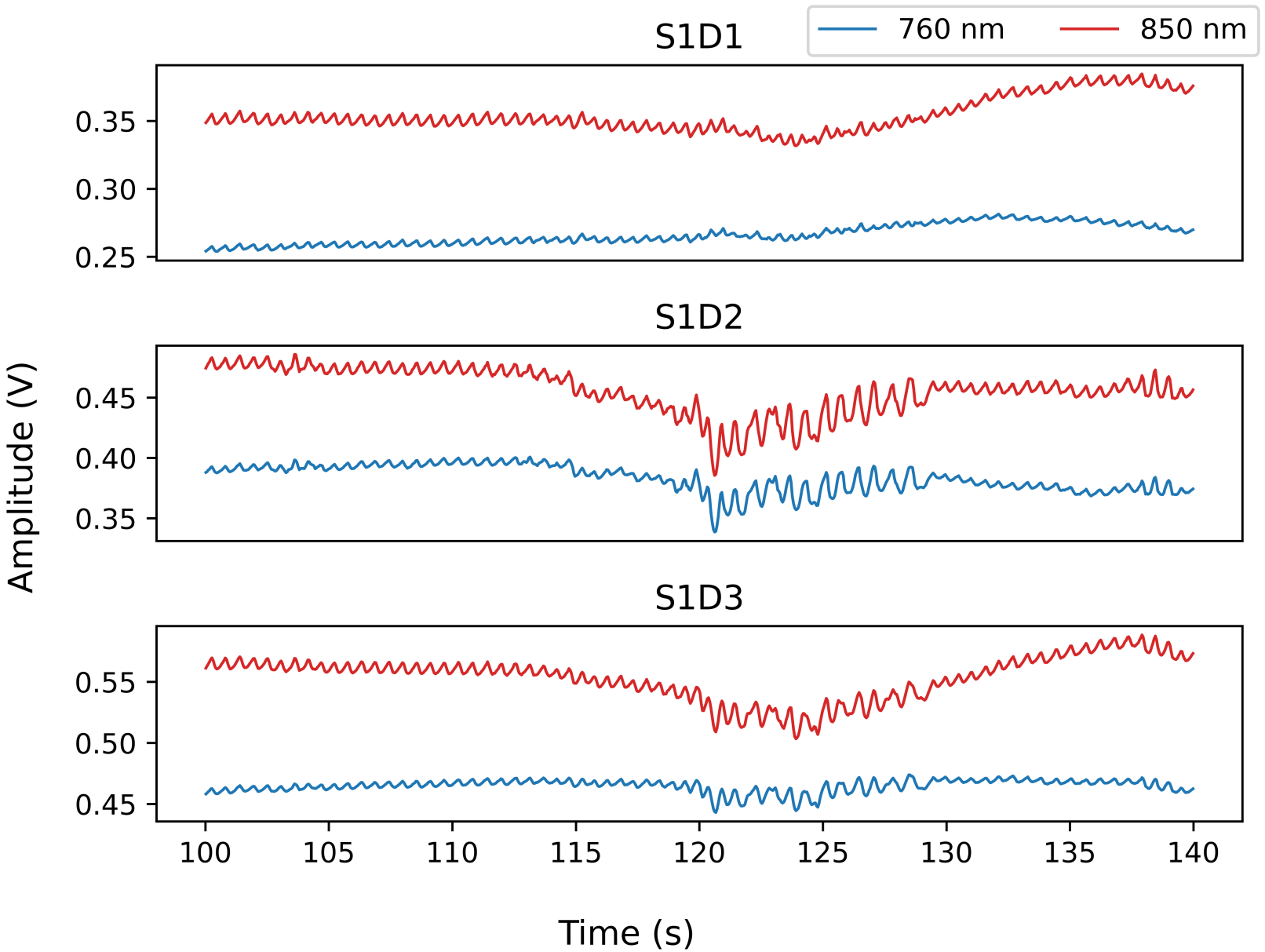
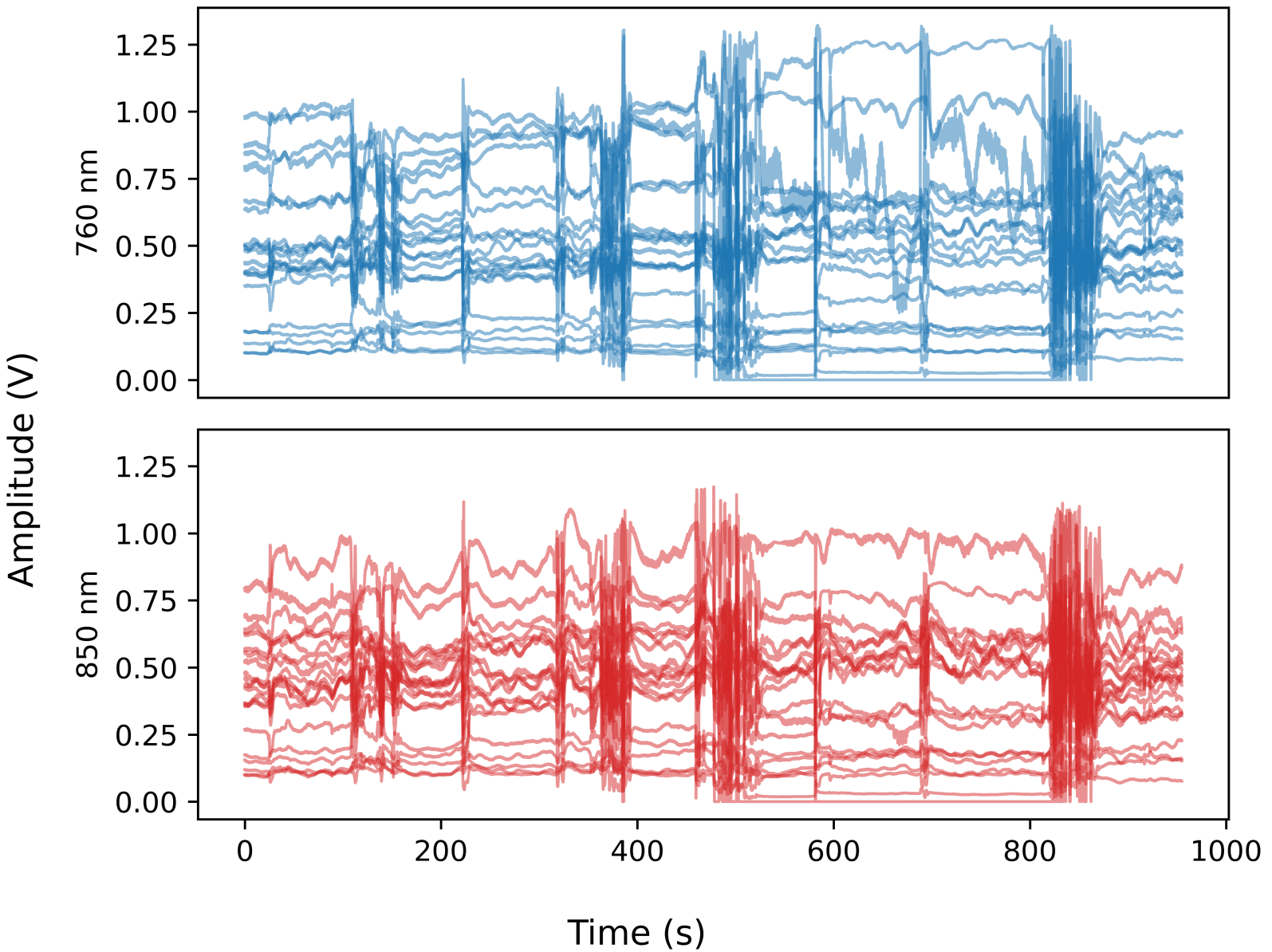


# The ugly: sleeping neonate



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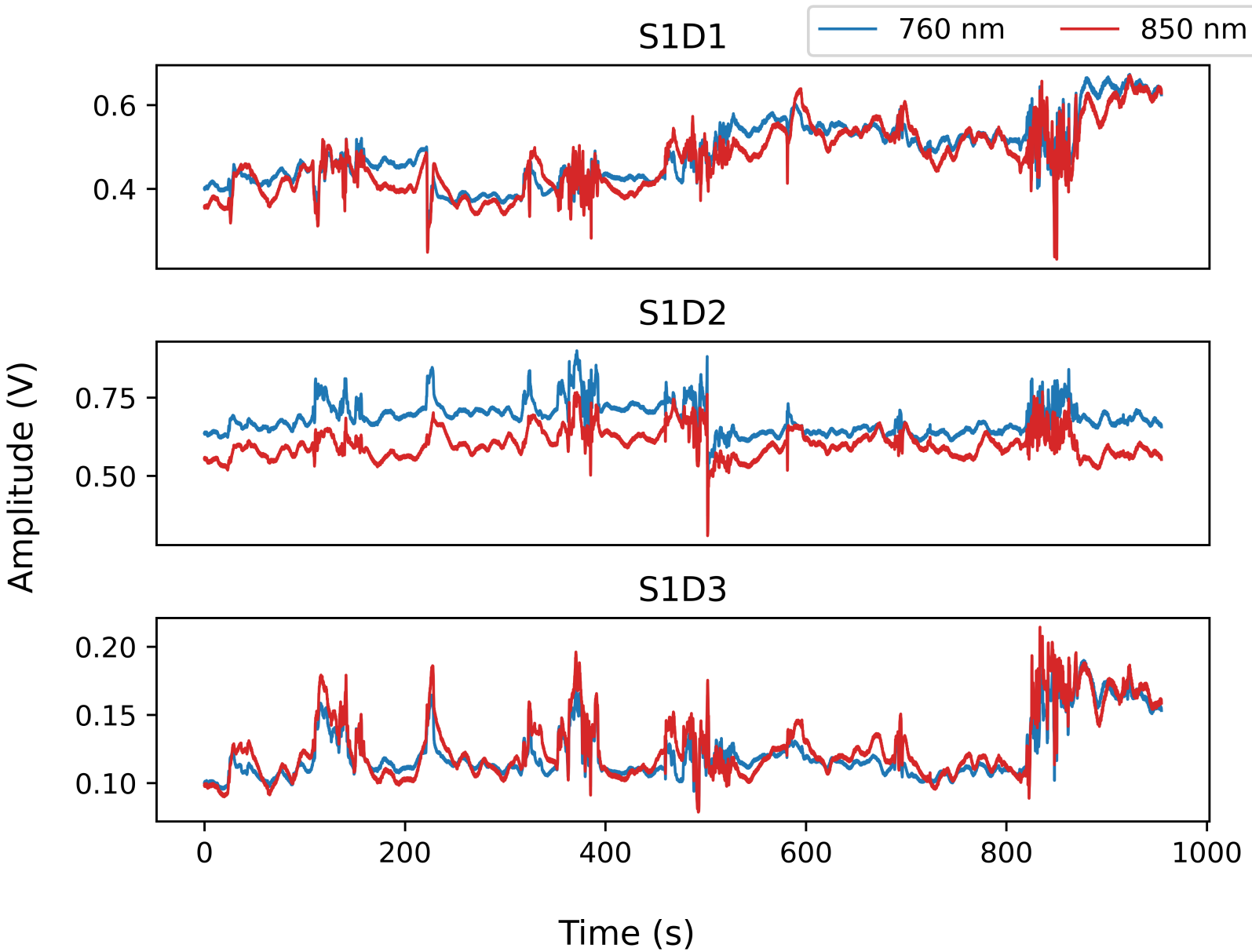

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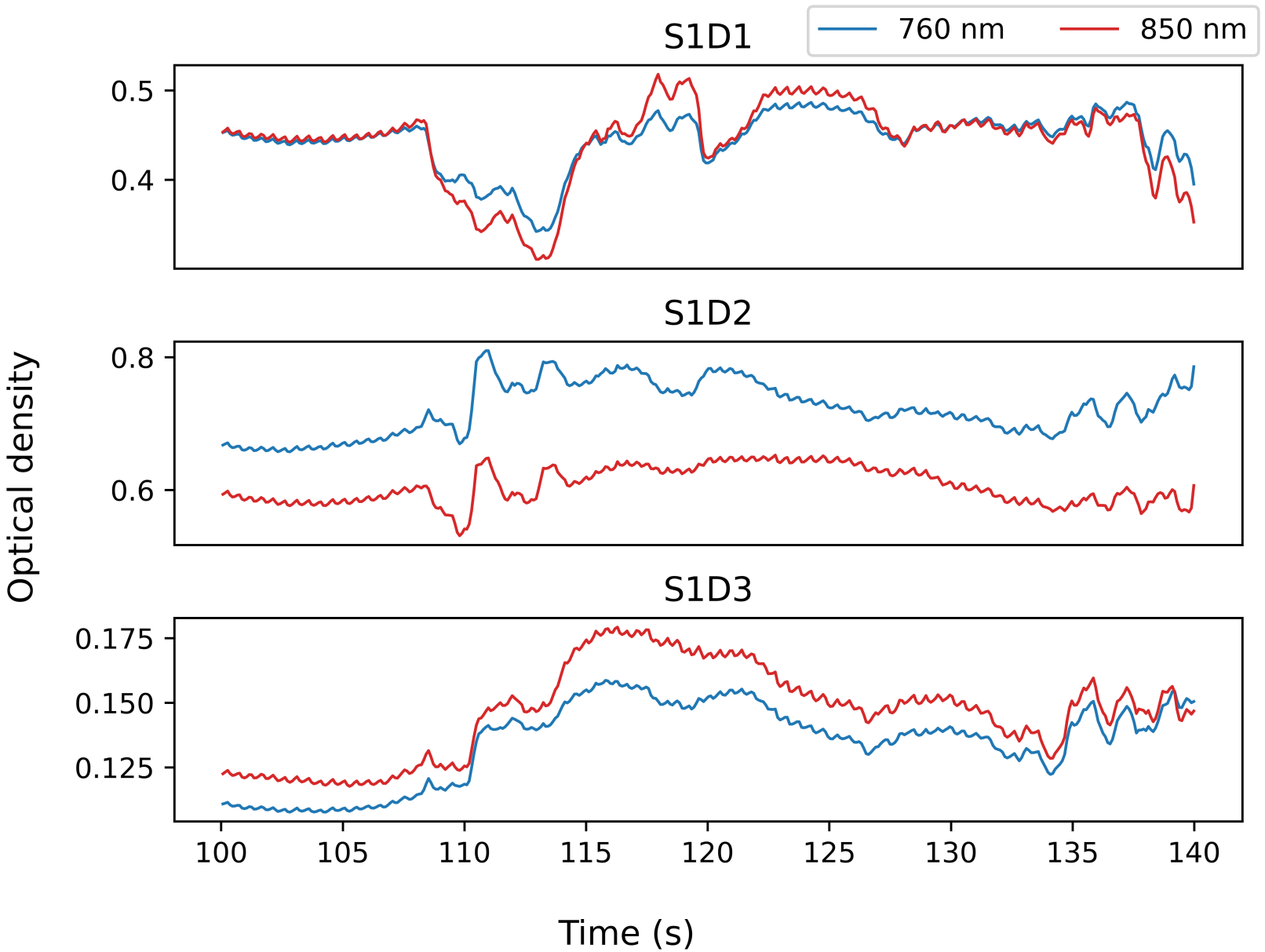


# Signal quality

## Before data collection

Calibration metrics:

- Coefficient of variation (CV)
- Signal-to-noise ratio (SNR)
- Dark light

## After data collection

- Signal-to-noise ratio (SNR)
- **Scalp coupling index (SCI)**
- **Peak spectral power (PSP)**

Can be used before data collection too!

(Pollonini, Bortfeld, and Oghalai 2016)

# Signal quality: after data collection

**Scalp coupling index (SCI)**: Index of optode coupling with scalp based on cardiac pulse (Pollonini et al. 2014)

1. Band-pass filter the signal in the heartbeat frequency of interest
2. Normalize time series and segment into rolling windows (e.g., 5 seconds)
3. For each window: Zero-lag cross-correlation between time series in both wavelengths
4. Compare against threshold

$$\mathrm{SCI} = \rho_{X_{\lambda_1} X_{\lambda_2}}(t=0) = \frac{X_{\lambda_1 t} - \bar{X}_{\lambda_1}, X_{\lambda_2 t} - \bar{X}_{\lambda_2}}{\sigma_{\lambda_1} \sigma_{\lambda_2}}$$

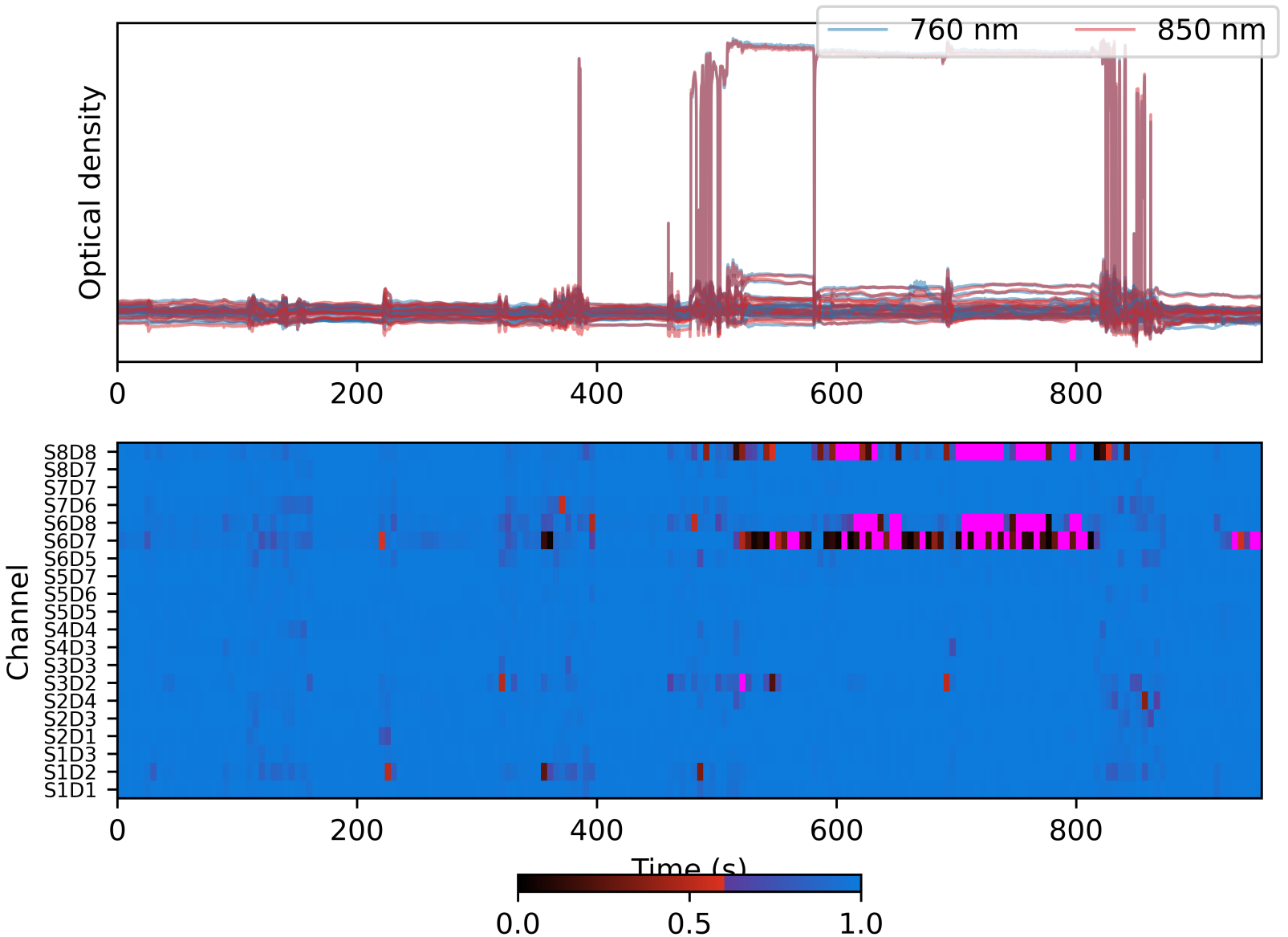
# Signal quality: after data collection

**Scalp coupling index (SCI)**: Index of optode coupling with scalp based on cardiac pulse (Pollonini et al. 2014)

```
from cedalion.nirs import cw
from cedalion.sigproc import quality

od = cw.int2od(rec["amp"])
window_length = 5 * units.s
sci, sci_mask = quality.sci(od, window_length, 0.8)
```

# Signal quality: after data collection



# Signal quality: after data collection

**Scalp coupling index (SCI)**: Index of optode coupling with scalp based on cardiac pulse (Pollonini et al. 2014)

*Limitation*: Motion arctifacts can inflate SCI (synchronised, high-amplitude perturbations in both signals)

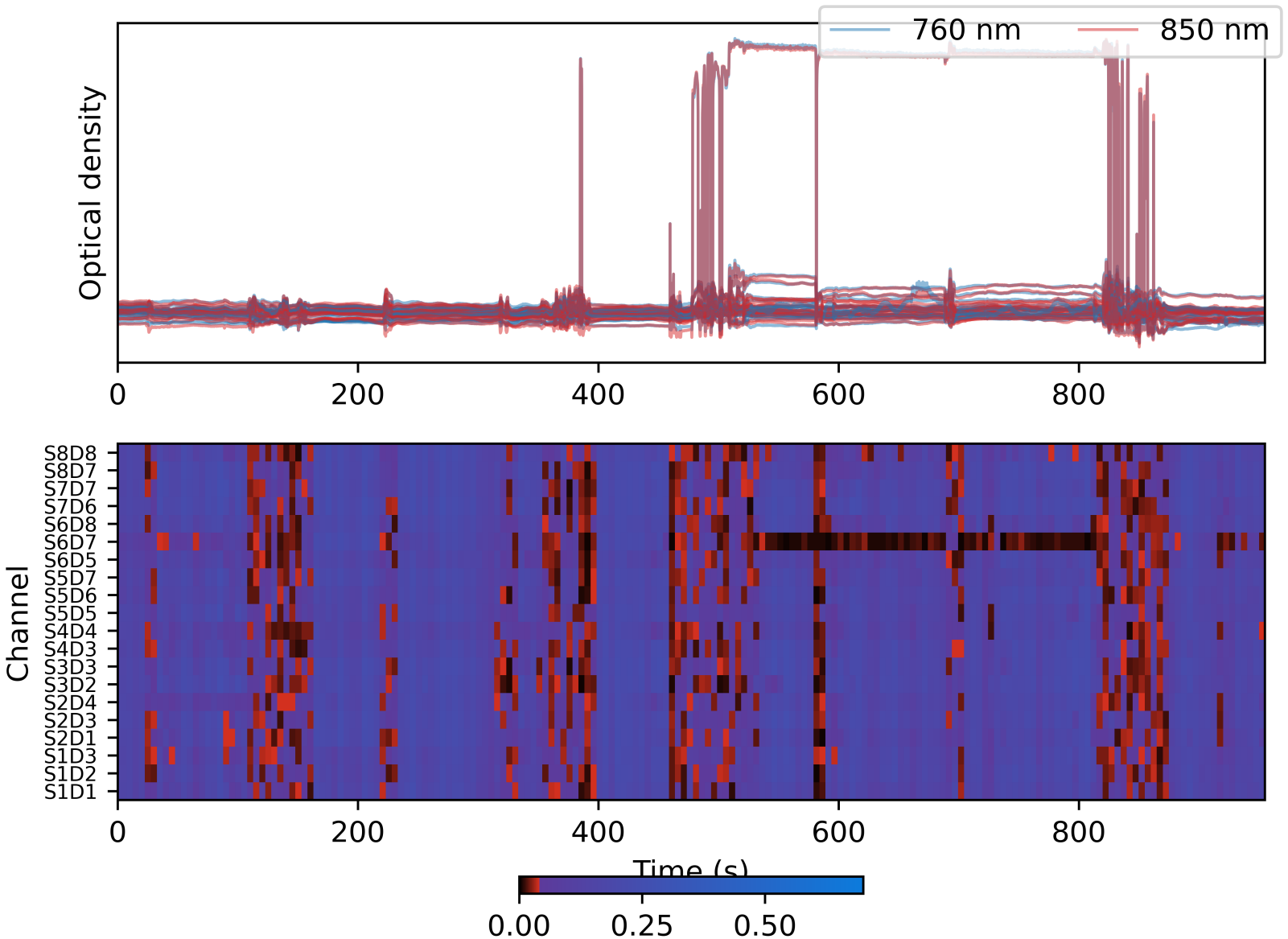
# Signal quality: after data collection

**Peak spectral power (PSP)**: peak power of cross-correlation between both wavelengths in the hearbeat frequency range.

1. Same as in SCI, now considering multiple lags (instead of just zero-lag)
2. Get peak power across lags
3. Compare against threshold

Recommended in combination with SCI (Pollonini, Bortfeld, and Oghalai 2016)

# Signal quality: after data collection



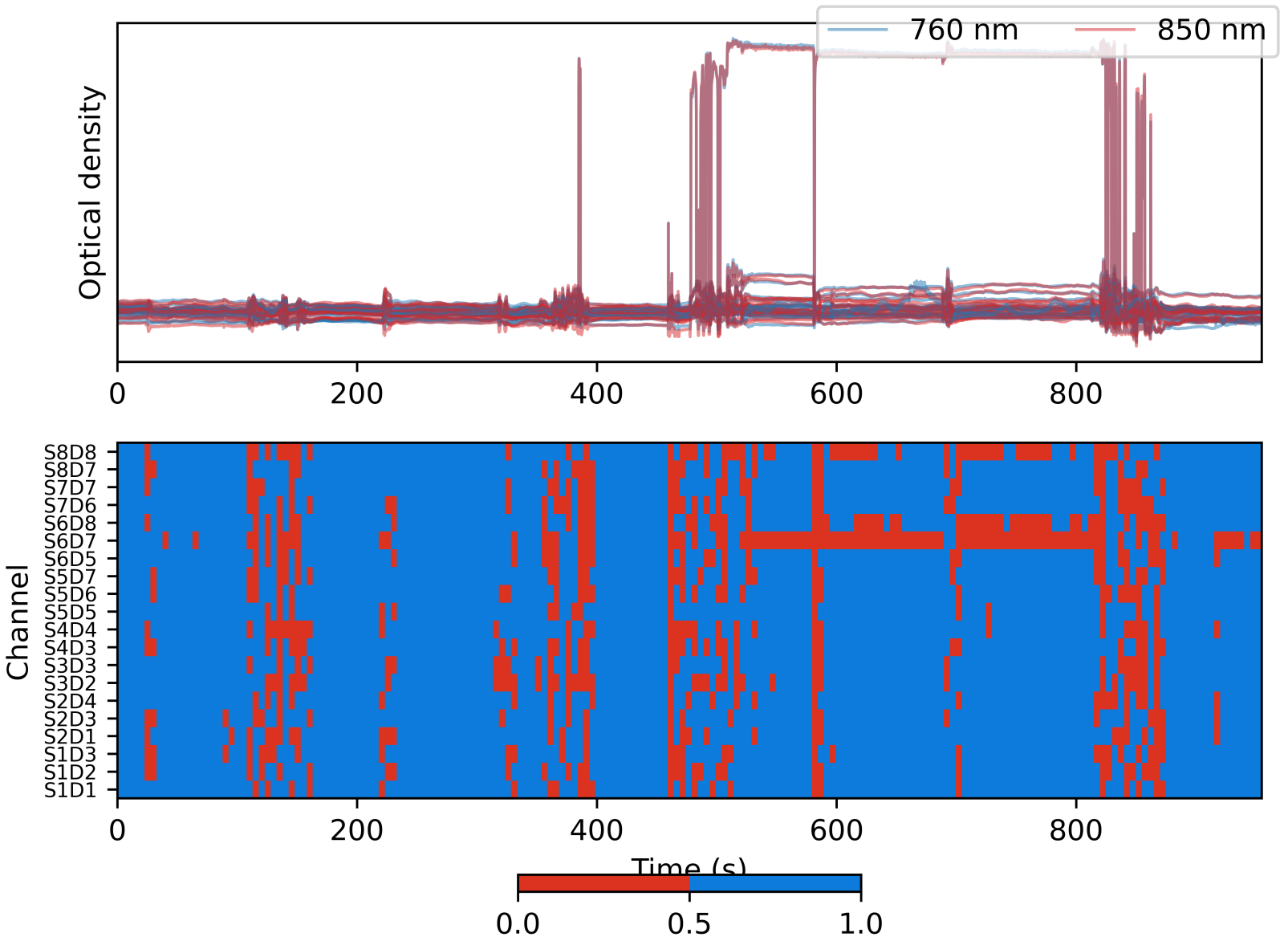
# Signal quality: choosing thresholds

**Sample size vs. data quality trade-off.**

Recommendations:

- Adults: *SCI*=0.8, *PSP*=0.1 (Pollonini, Bortfeld, and Oghalai 2016)
- Infants: *SCI*=0.6, *PSP*=0.04-0.05 (Beaton et al. 2026)

# Signal quality: choosing thresholds



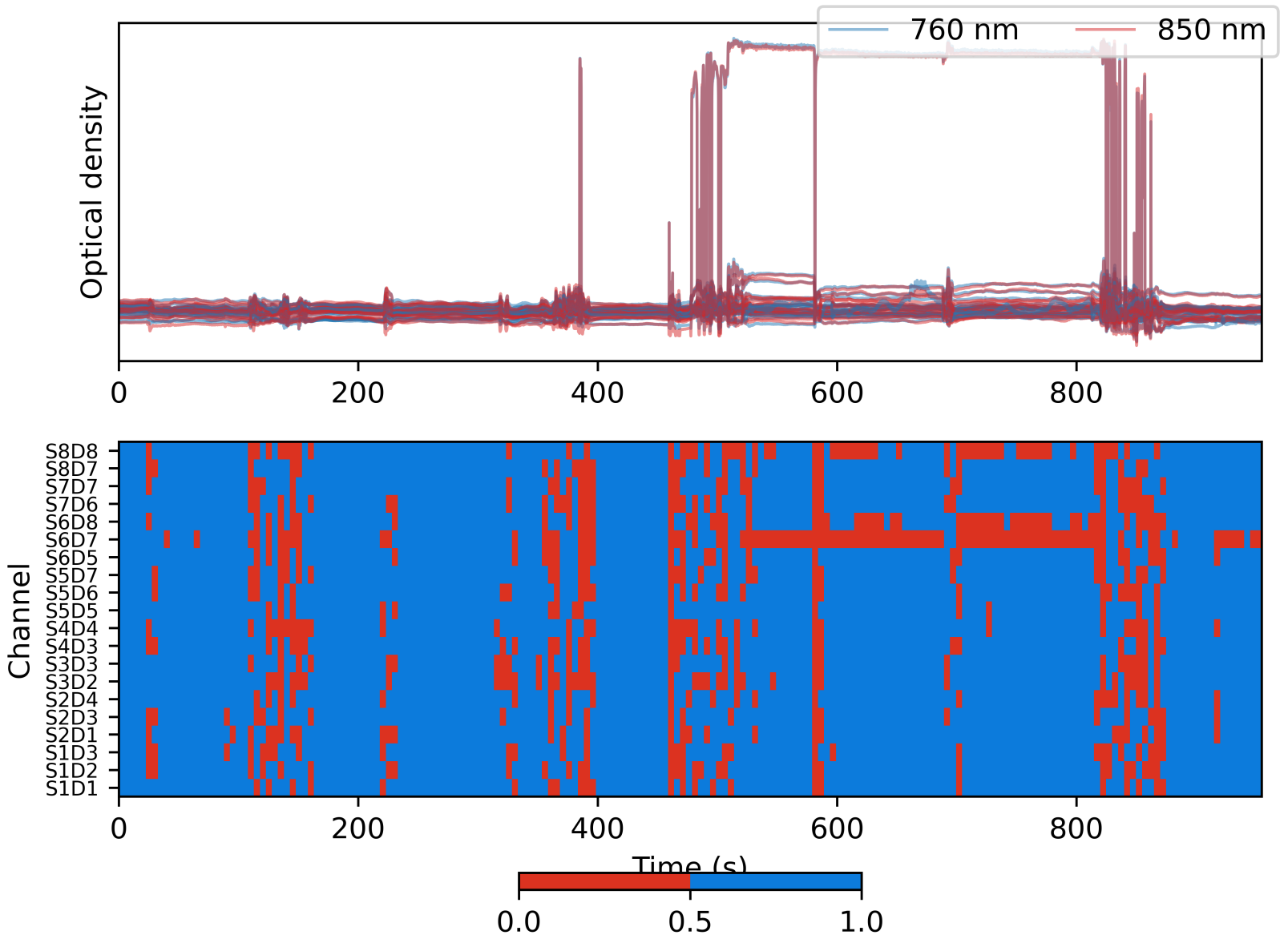
# Signal quality: channel pruning

Channels with less than some threshold proportion of above-threshold windows are excluded from further analyses (e.g., 0.7).

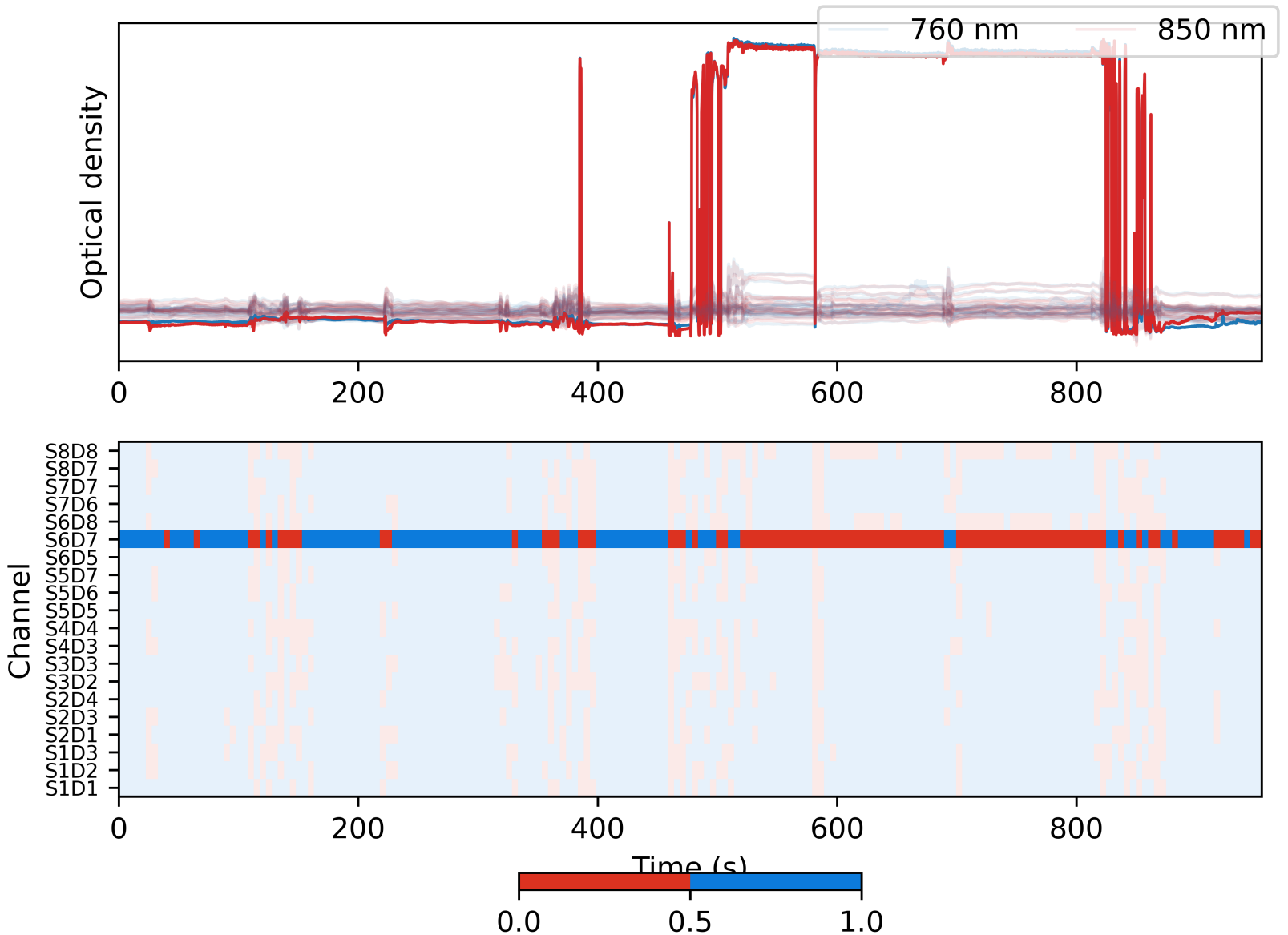
```
quality_mask = [time_clean >= 0.7]
rec["od"], exc_ch = quality.prune_ch(rec["od"], quality_mask, "all")
od_pruned, drop_list = quality.prune_ch(rec["od"], masks, "all")
```

Channel interpolation? Work in progress.

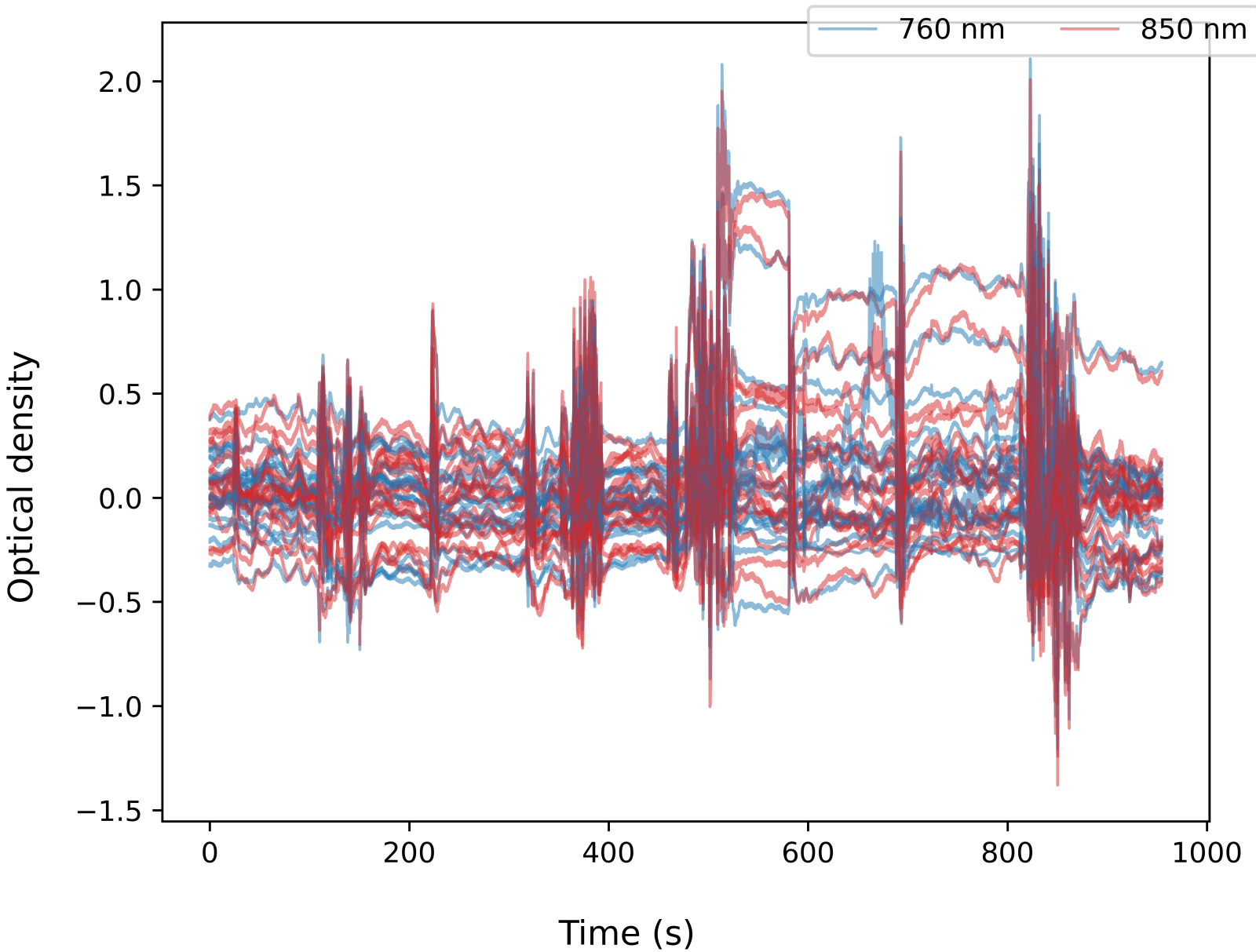
# Signal quality: channel pruning



# Signal quality: channel pruning



# Signal quality: channel pruning



# Motion arctifact attenuation

Motion arctifacts (MA) are caused my changes optode position due to movement (cap slides around or separates from scalp).

Brigadoi et al. (2014):

- **Spikes**: fast, high amplitude changes
- **Baseline shifts**
- **Drift**: low frequency variations

# Motion arctifact attenuation

## External signals

- Short channel regression*
  - But see Emberson et al. (2016)
- Accelerometer (Cui et al. 2015)

## Signal correction methods

- **Wavelet filtering** (Molavi and Dumont 2012)
- **Temporal deivative distribution repair (TDDR)** (Fishburn et al. 2019)
- Savitzky-Golay spline interpolation (Jahani et al. 2018)
- Principal component analysis (PCA)

# Motion arctifact attenuation

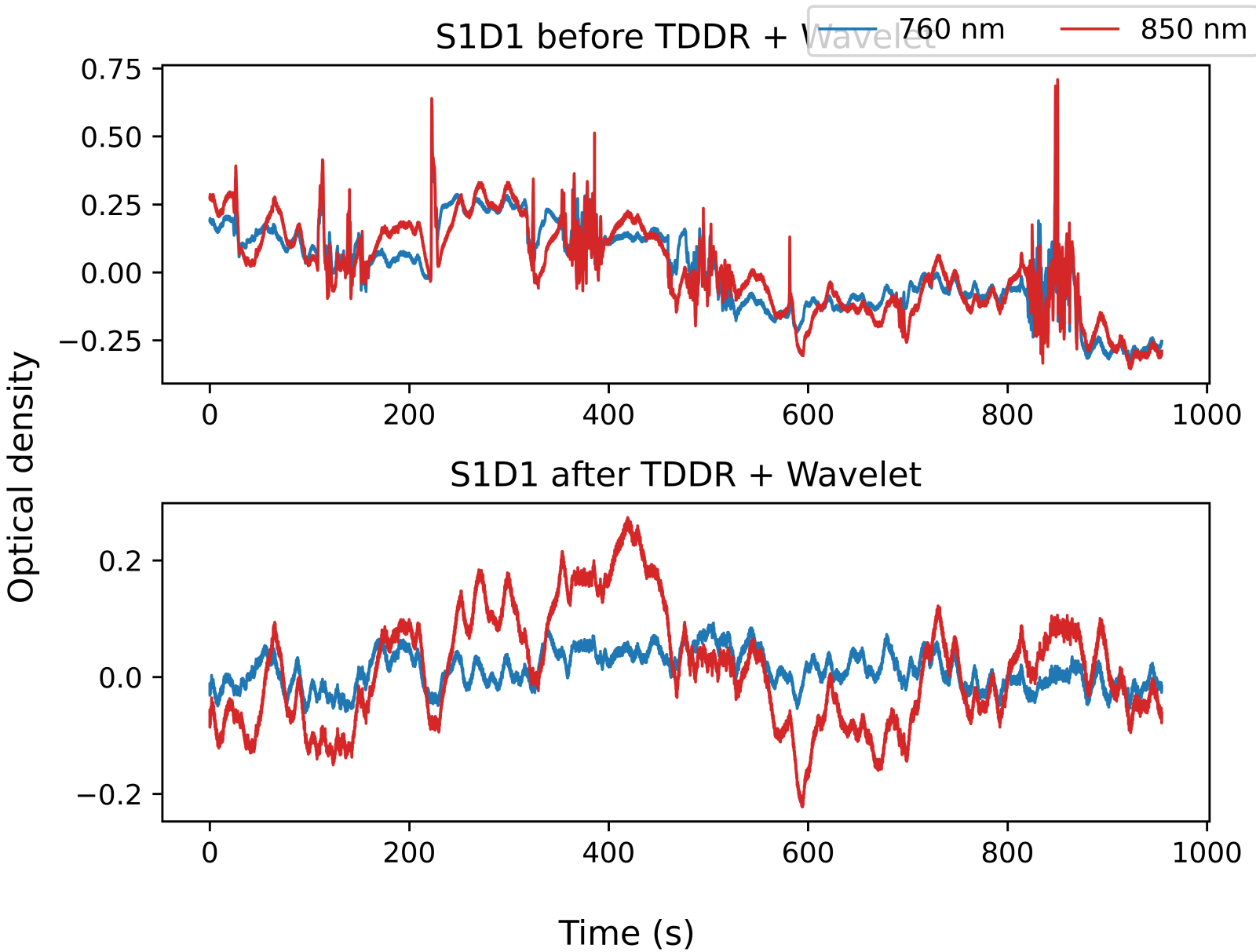
Current recommendation for pediatric data: combination of **wavelet filtering** and **spline interpolation** (Di Lorenzo et al. 2019).

We found **TDDR + wavelet** performed best in neonatal datasets (in-lab annecdotical data).

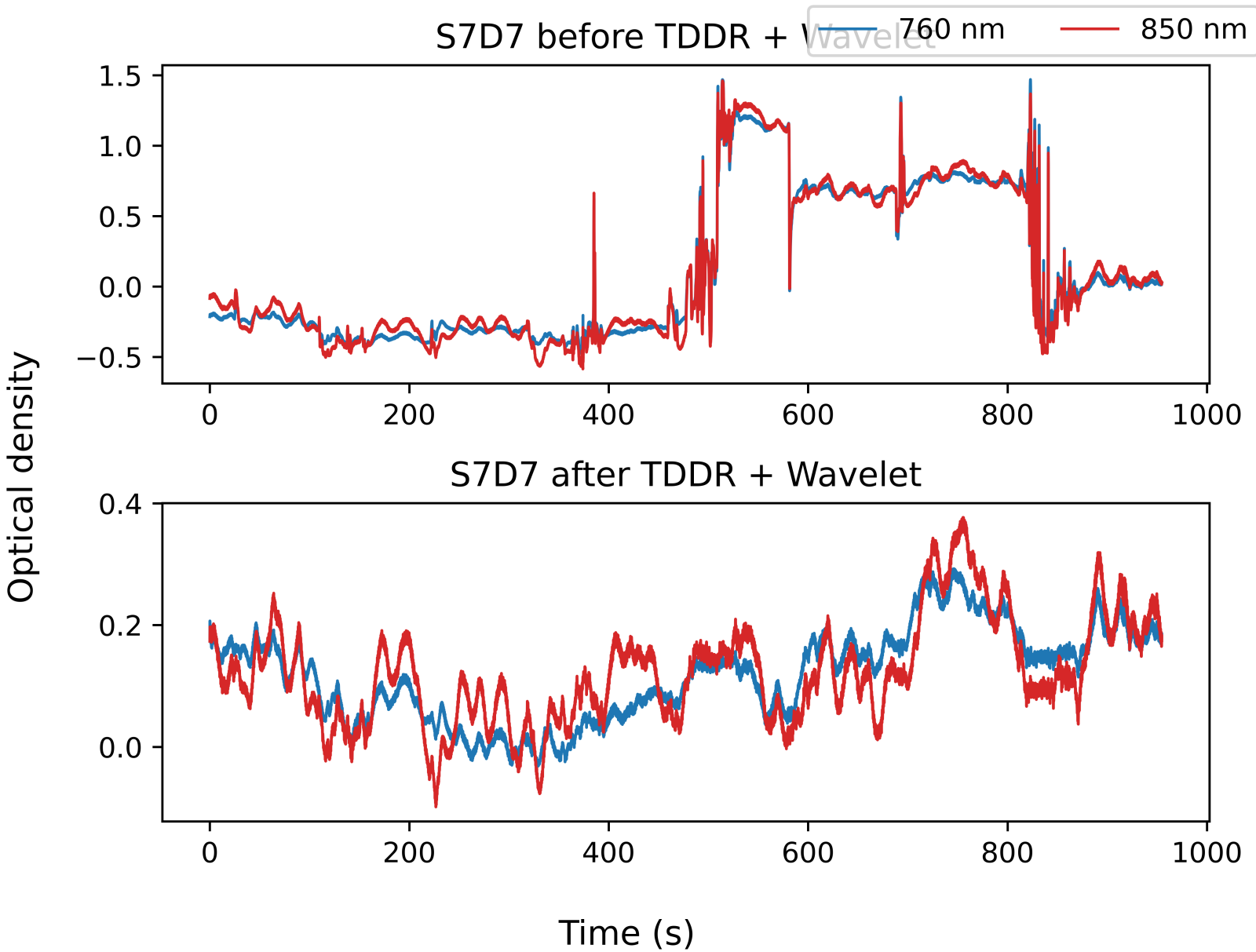
```
from cedalion.sigproc import motion_correct

od = motion_correct.tddr(motion_correct.wavelet(od))
```

# Motion arctifact attenuation



# Motion arctifact attenuation



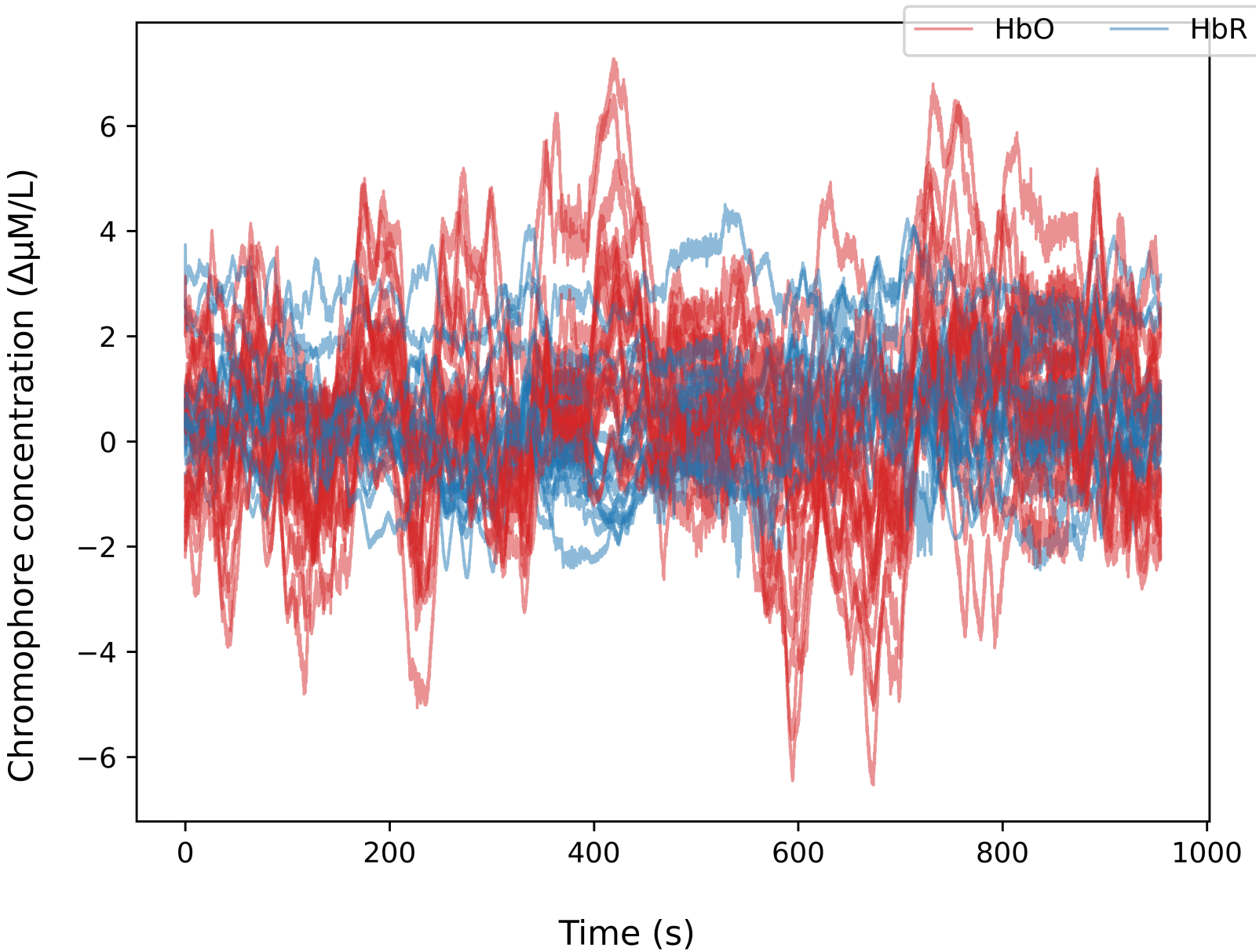
# Concentration of HbO and HbR

```
dpf_vals = [4.2, 5.3]
dpf_coords = {"wavelength": rec["od"].wavelength}
dpf = xr.DataArray(dpf_vals, dims="wavelength", coords=dpf_coords)
rec["conc"] = cw.od2conc(rec["od"], rec.geo3d, dpf)
```

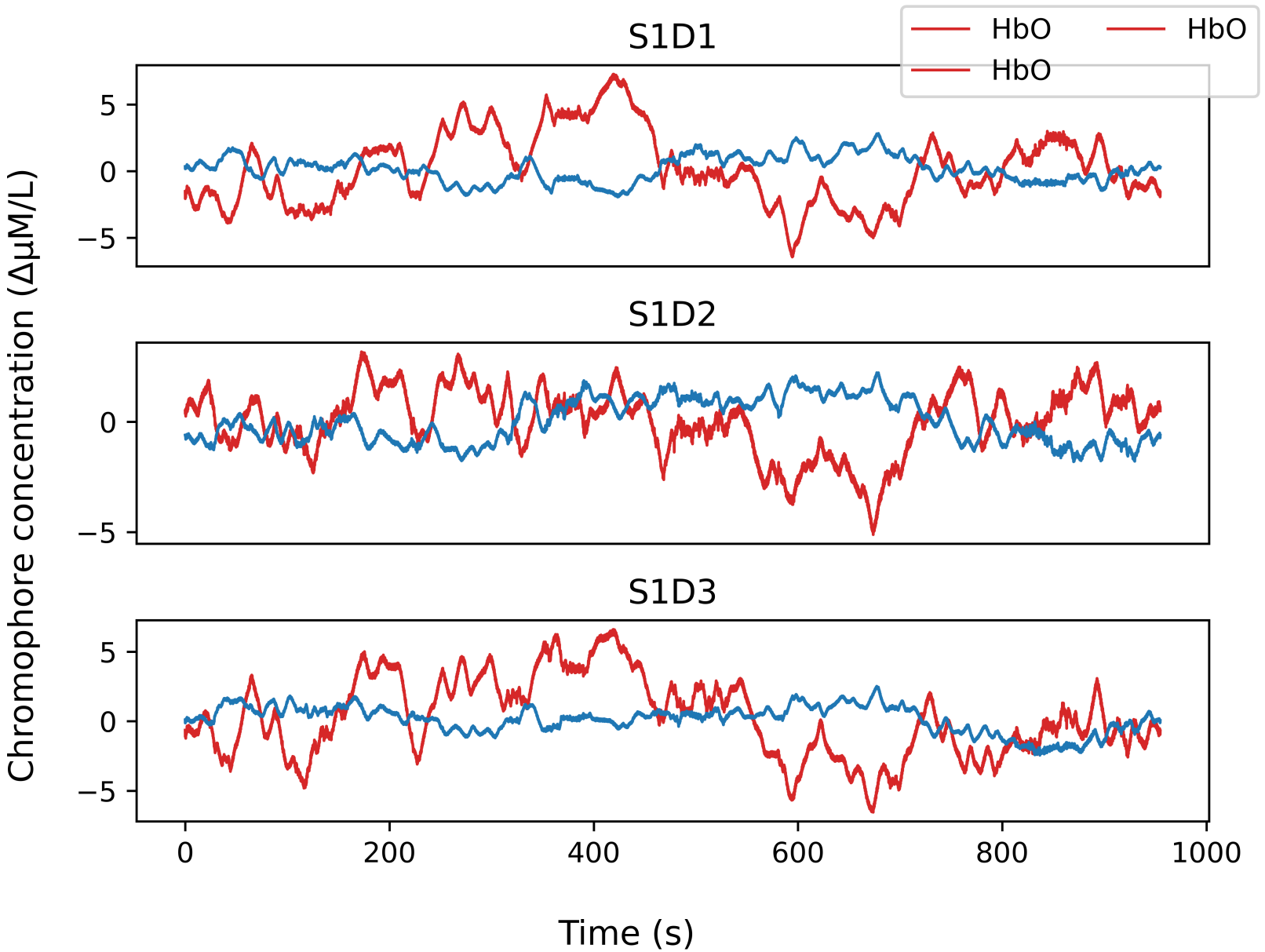
**Differential pathlength factor** ($\overline{DPF}$)

Constant in the mBLL equation. Depends on the characteristics of the underlying biological tissue; choose age appropriate values.

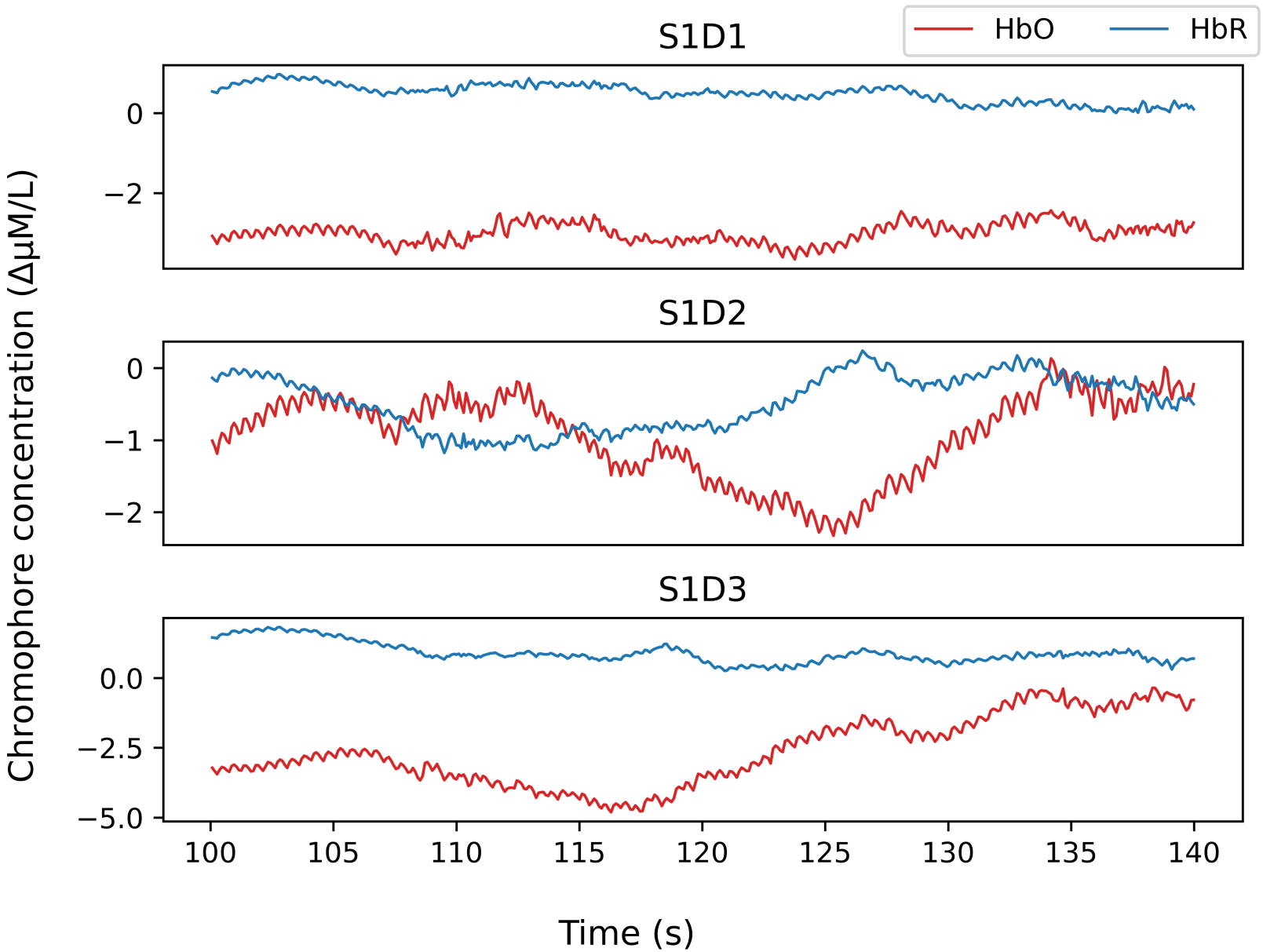
# Concentration of HbO and HbR



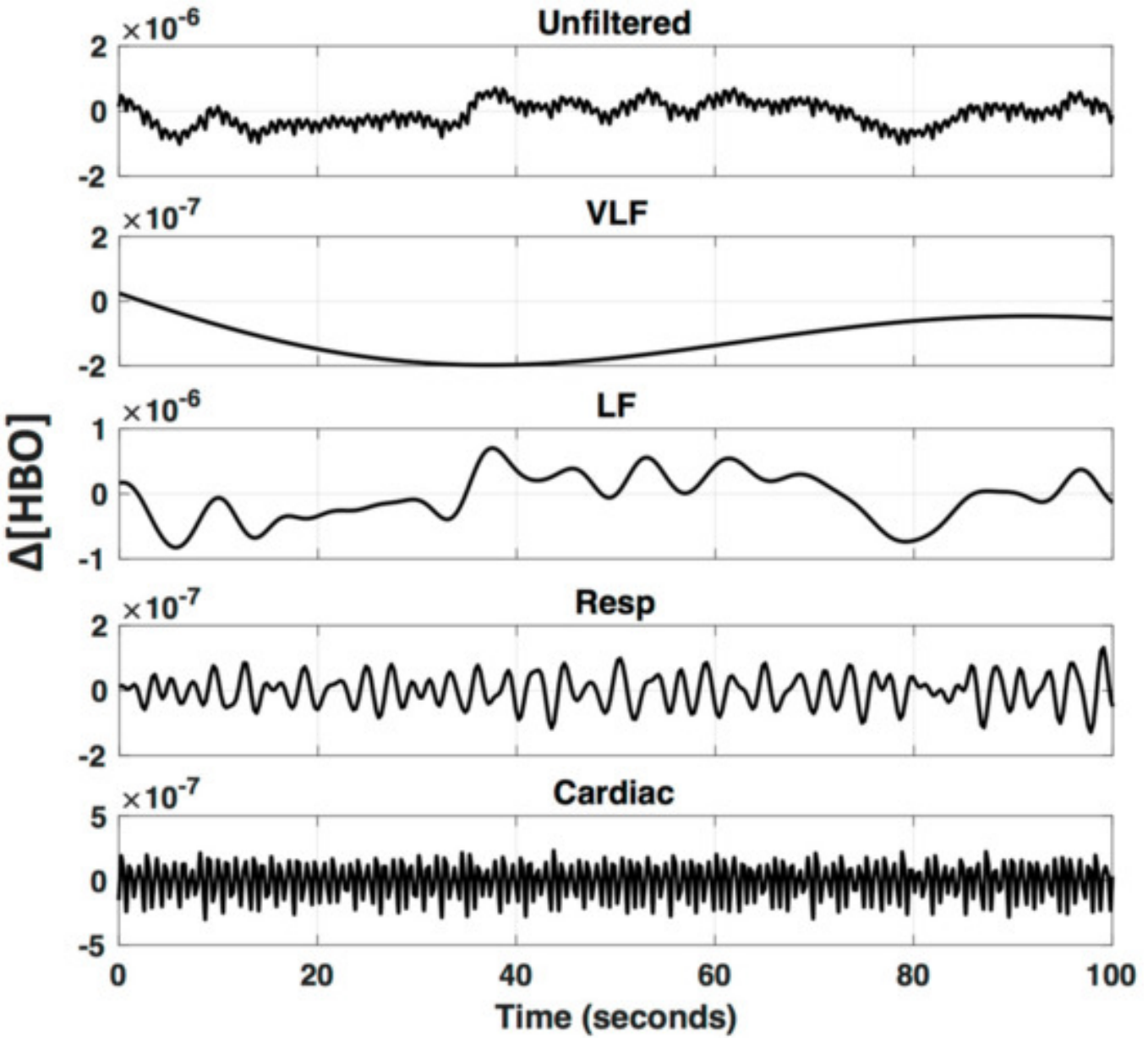
# Concentration of HbO and HbR



# Concentration of HbO and HbR



# Physiological arctifact attenuation: frequency filtering

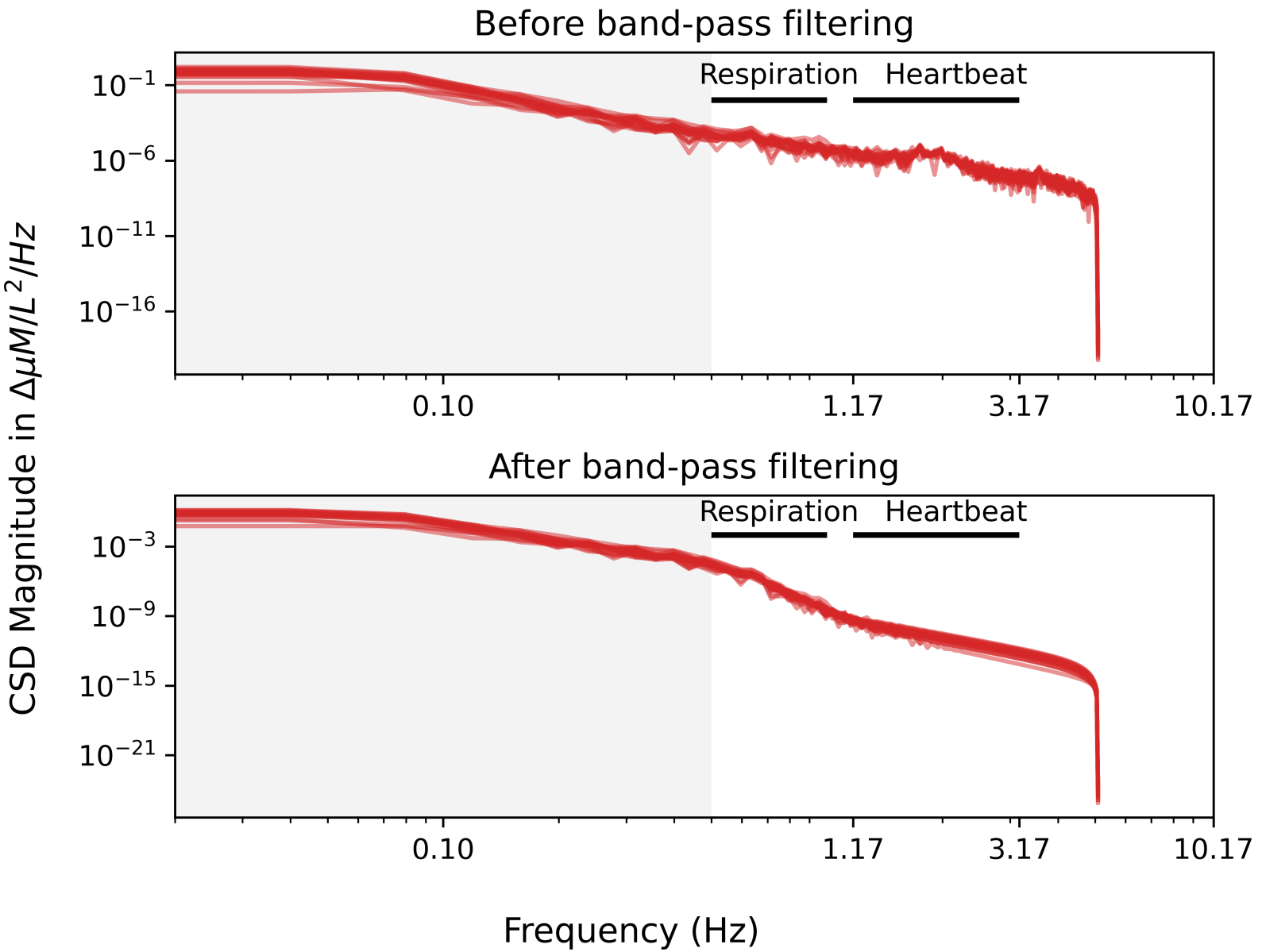


# Physiological arctifact attenuation: frequency filtering

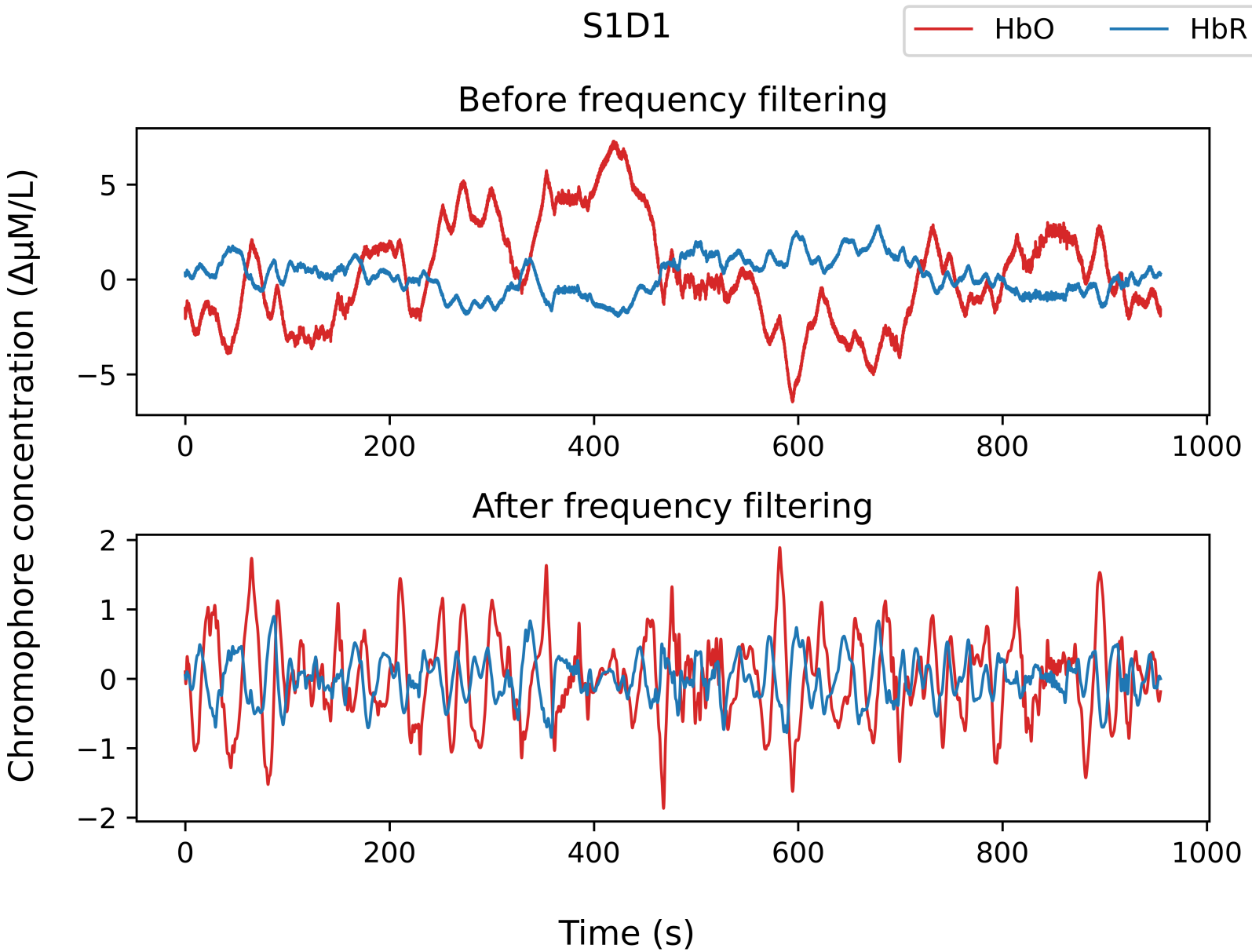
| | Heartbeat | Respiration | Mayer waves | Very low frequency waves |
|---|---|---|---|---|
| Adults | 1 Hz | 0.3 Hz | 0.1 Hz | < 0.04 Hz |
| Infants | 1.17-3.17 Hz | 0.5-1 Hz | Unknown | Unknown |

```
fmin, fmax = 0.02, 0.5
rec["conc"] = rec["conc"].cd.freq_filter(
    fmin=fmin * units.Hz,
    fmax=fmax * units.Hz,
)
```

# Physiological arctifact attenuation: frequency filtering



# Physiological arctifact attenuation: frequency filtering



# Physiological arctifact attenuation: frequency filtering

# Physiological arctifact attenuation: frequency filtering

# Physiological arctifact attenuation: frequency filtering

# Epoching and averaging

```
1 print(rec.stim)
```

```
         onset  duration  value trial_type
25   10.518528      10.0    1.0          A
26  185.696256      10.0    1.0          A
27  232.783872      10.0    1.0          A
28  272.891904      10.0    1.0          A
29  309.067776      10.0    1.0          A
30  346.324992      10.0    1.0          A
31  438.730752      10.0    1.0          A
32  530.546688      10.0    1.0          A
33  737.574912      10.0    1.0          A
34  876.380160      10.0    1.0          A
35   59.670528      10.0    1.0          B
36  108.036096      10.0    1.0          B
37  148.242432      10.0    1.0          B
```

# Epoching and averaging

# Epoching and averaging

# Epoching and averaging

```
epochs = rec["conc"].cd.to_epochs(
    rec.stim,  # stimulus dataframe
    ["FTapping/Left", "FTapping/Right"],  # select fingertapping events, di
    before=5 * units.s,  # seconds before stimulus
    after=30 * units.s,  # seconds after stimulus
)
# calculate baseline
baseline = epochs.sel(reltime=(epochs.reltime < 0)).mean("reltime")

# subtract baseline
epochs =- baseline
epochs = epochs - baseline
epochs = nirs.detrend_epochs(epochs)

# average
hrf = epochs.groupby("trial_type").mean("epoch")
```

# Epoching and averaging

# Skin and hair type

Yücel et al. (2025):

- Fitzpatrick Skin Phototype Scale (I, II, III, IV)
- Fischer-Saller Scale, Andre Walker Hair Typing System, and the FIA Hair Typing System

# Functional Near-Infrared Spectroscopy

If we shine near-infrared light into biological tissue, we can estimate **changes in relative concentration** of HbO and HbR time (**functional**).

**modified Beer-Lambert Law (mBLL)**

$$\Delta A(\lambda_1 = 760\ nm) = \mathrm{DPF} \cdot r \cdot [\epsilon_{HbO}(\lambda_1)\Delta c_{HbO} + \epsilon_{HbO}(\lambda_1)\Delta c_{HbO}]$$
$$\Delta A(\lambda_2 = 850\ nm) = \mathrm{DPF} \cdot r \cdot [\epsilon_{HbO}(\lambda_2)\Delta c_{HbR} + \epsilon_{HbR}(\lambda_2)\Delta c_{HbR}]$$

# Thank you for you attention!

# Data organisation: BIDS format

bids.neuroimaging.io

- Proposed **standard** for **neuroimaging** datasets (EEG, fNIRS, MRI, etc.)
- Adopted and encouraged by the **Society for fNIRS** (SFNIRS)
- Increases **reproducibility**, transparency, and documentation
- Ready to share in **BIDS-compliant** repositories (e.g., OpenNeuro.)

# Data organisation: BIDS format

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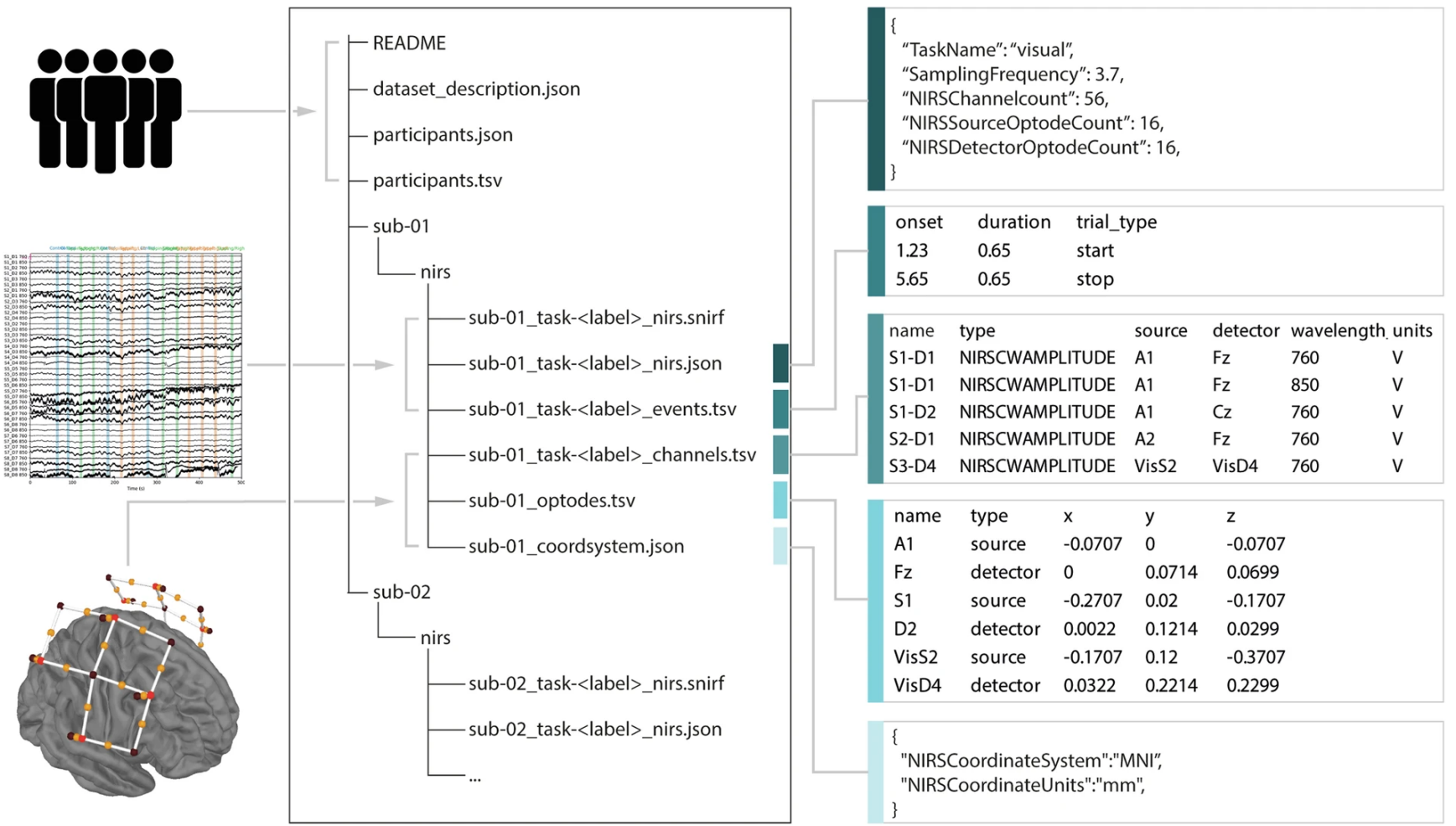

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